

An isomorphism between the Hopf algebras $\mathfrak{A}$ and $\mathfrak{B}$ of Jacobi diagrams in the theory of knot invariants

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Abstract

We construct a graded Hopf algebra $\mathfrak{B}$ from the symmetric algebra of a metrized Lie algebra, and examine the structure of low-dimensional spaces of the grading. With this we construct and diagrammise an isomorphism to the Hopf algebra $\mathfrak{A}$, this algebra arising from the universal enveloping algebra of the same metrized Lie algebra.

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0 Motivating remarks

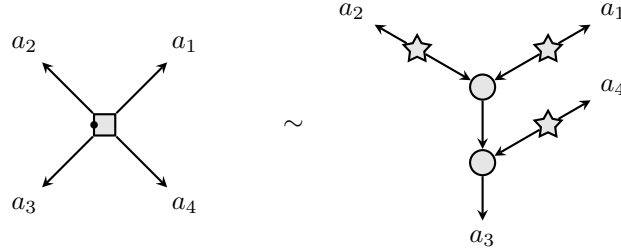
Penrose's diagrammatic analogue of tensor calculus was applied to generate a diagrammatic construction of the universal enveloping algebra $\mathfrak{U}(\mathfrak{g})$ of a metrized Lie algebra $\mathfrak{g}$. The Poincare-Birkhoff-Witt theorem gives an isomorphism between $\mathfrak{U}(\mathfrak{g})$ and $\mathfrak{S}(\mathfrak{g})$, the symmetric algebra of the same metrized Lie algebra. It is therefore of potential interest to construct a diagrammization of $\mathfrak{S}(\mathfrak{g})$.

0.1 Diagrammisation of $\mathfrak{U}(\mathfrak{g})$ through the algebra $\mathcal{A}$

Previously (see [4]) $\mathfrak{U}(\mathfrak{g})$ was diagrammised, for $\mathfrak{g}$ with a non-degenerate symmetric bilinear form and an orthonormal basis. The algebra was generated by disjoint union, linear combination, and contraction, from the basis elements

$$\left\{ \begin{array}{c} \begin{array}{c} i \\ \downarrow \\ \star \\ \uparrow \\ j \end{array}, \quad \begin{array}{c} i \\ \uparrow \\ \star \\ \downarrow \\ j \end{array}, \quad \begin{array}{c} i \quad j \\ \searrow \quad \swarrow \\ \circ \\ \downarrow \\ k \end{array}, \quad \begin{array}{c} a_1 \quad a_2 \\ \swarrow \quad \searrow \\ \square \\ \downarrow \\ a_p \end{array} \quad \forall p \in \mathbb{N} \end{array} \right\} \quad (0.1)$$

The third element in this list is termed the diagram of the Lie bracket (or simply the Lie diagram). The last element in this list describes an anchored tensor with p arrows directed outward. Through isomorphism in a common vector space, it may be constructed by a combination of the second and third diagrams, for example,



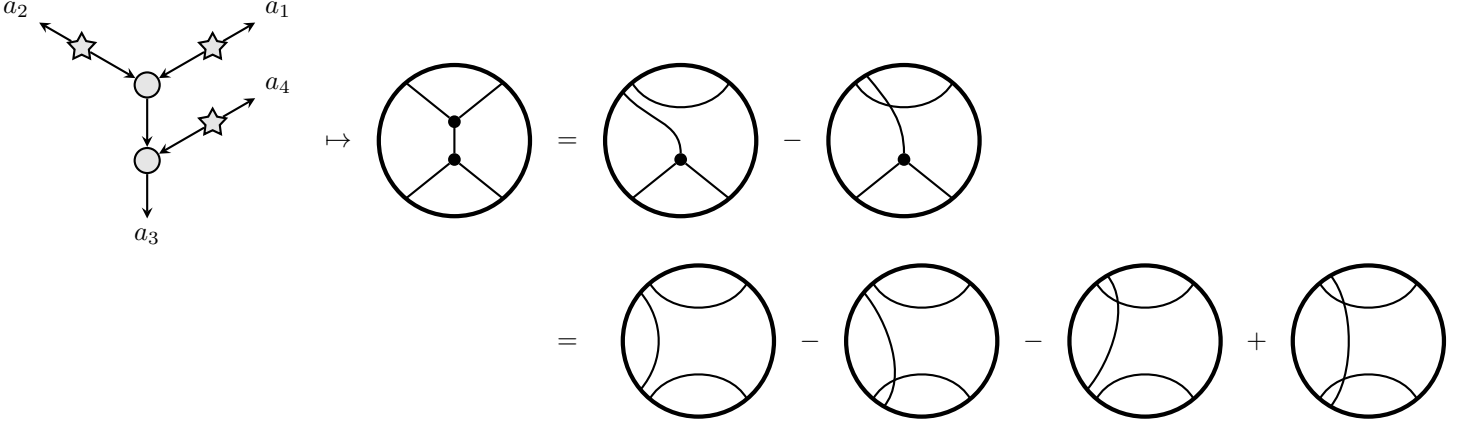
Hence the anchors may be disregarded¹. The basis of the described algebra was then reduced to four elements, namely the first three of (0.1) and $\square \rightarrow i$. As the next step was to attach a skeleton to the outward edges of the diagrams, the first two diagrams of (0.1) became irrelevant, as they only reverse direction of edges.

The algebra $\mathcal{A}$ was then constructed (see [3]) using the following basis modulo a skeletal version of the simplification of the Lie diagram,

$$\left\{ \begin{array}{c} \begin{array}{c} i \quad j \\ \searrow \quad \swarrow \\ \circ \\ \downarrow \\ k \end{array}, \quad \begin{array}{c} \square \\ \downarrow \\ i \end{array} \end{array} \right\} \quad \begin{array}{c} i \quad j \\ \searrow \quad \swarrow \\ \circ \\ \downarrow \\ k \end{array} = \begin{array}{c} i \quad j \\ \downarrow \quad \downarrow \\ i \quad j \end{array} - \begin{array}{c} i \quad j \\ \swarrow \quad \searrow \\ i \quad j \end{array}$$

In $\mathcal{A}$, directions and labels are ignored, and as there is only one type of vertex not on the skeleton, the previously used different shapes are no longer used. For example,

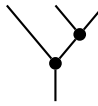
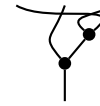
¹see [4] Sec. 3.6.3 for a complete argument



The link with chord diagrams and knot invariants is now clear. Here we first focus on diagrammising $\mathfrak{S}(\mathfrak{g})$ and extracting the algebra $\mathcal{B}$, and in Section 3 the algebra $\mathcal{A}$ described above will be associated by an isomorphism to $\mathcal{B}$, through their Hopf algebra structures.

0.2 Diagrammisation of $\mathfrak{S}(\mathfrak{g})$

Penrose's tensor calculus was applied to the construction of the symmetric algebra $\mathfrak{S}(\mathfrak{g})$ in [4], the key properties being no anchors or labels. As in $\mathcal{A}$ edges in the diagrammisation of $\mathfrak{S}(\mathfrak{g})$ will be undirected as well. Hence $\mathfrak{B}$ may be described as the space of linear combinations of unitriangular diagrams, not necessarily connected, modulo the two relations:

AS relation:  = -  Jacobi relation:  +  +  = 0

We rearrange the Jacobi relation and apply the AS relation to get an equivalent statement, which from now on will be termed the IHX relation.

$$\begin{array}{lcl}
 \begin{array}{c} c \quad b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ | \\ d \end{array} & \rightarrow & \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ \diagup \quad \diagdown \\ c \quad d \end{array} \\
 \begin{array}{c} c \quad b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ | \\ d \end{array} & \rightarrow & \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ | \\ d \end{array} = - \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ \diagup \quad \diagdown \\ c \quad d \end{array} = - \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ \diagup \quad \diagdown \\ c \quad d \end{array} \\
 \begin{array}{c} c \quad b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ | \\ d \end{array} & \rightarrow & \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ | \\ d \end{array} = - \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ \diagup \quad \diagdown \\ c \quad d \end{array} = \begin{array}{c} b \quad a \\ \diagdown \quad \diagup \\ \bullet \\ \diagup \quad \diagdown \\ c \quad d \end{array}
 \end{array}$$

The edges have been marked to keep track of the manipulations. The above rotations and rearrangements of the pendent edges lead to a parallel diagram equality. Hence the Jacobi relation is equivalent to:

$$\begin{array}{c} \text{Y-junction} \\ \text{with top edge marked} \end{array} - \begin{array}{c} \text{X-junction} \\ \text{with top edge marked} \end{array} + \begin{array}{c} \text{X-junction} \\ \text{with bottom edge marked} \end{array} = 0 \quad \Longleftrightarrow \quad \begin{array}{c} \text{I-junction} \\ \text{with top edge marked} \end{array} = \begin{array}{c} \text{I-junction} \\ \text{with bottom edge marked} \end{array} - \begin{array}{c} \text{X-junction} \\ \text{with both edges marked} \end{array}$$

This is the IHX relation. We now begin with a description of the space of elements of $\mathfrak{B}$.

1 The vector space $\mathcal{B}$

1.1 Derivation

Definition 1.1.1. (The vector space $\mathcal{D}_m^o$) We define $\mathcal{D}_m^o$ for $m \in \mathbb{N}$ to be the space of all formal linear combinations of open Jacobi diagrams of degree m , or equivalently, connected univalent graphs of degree m with oriented vertices. For an open Jacobi diagram D , $\deg(D) = \frac{1}{2}|V(D)|$.

Definition 1.1.2. (The vector spaces $\mathcal{B}$ and $\mathcal{B}_m$) Define the vector spaces $\mathcal{B}$ and $\mathcal{B}_m$ by

$$\mathcal{B}_m = \frac{\mathcal{D}_m^o}{(\text{IHX}, \text{AS})} \quad \mathcal{B} = \bigoplus_{m \geq 0} \mathcal{B}_m \quad (1.1)$$

with the degree-preserving IHX, AS relations given by

$$\text{IHX: } \begin{array}{c} \text{I-junction} \\ \text{with top edge marked} \end{array} = \begin{array}{c} \text{I-junction} \\ \text{with bottom edge marked} \end{array} - \begin{array}{c} \text{X-junction} \\ \text{with both edges marked} \end{array} \quad \text{AS: } \begin{array}{c} \text{Y-junction} \\ \text{with top edge marked} \end{array} = - \begin{array}{c} \text{Y-junction} \\ \text{with bottom edge marked} \end{array} \quad (1.2)$$

Coefficients are taken over $\mathbb{C}$, the zero element is $\emptyset$ (termed the null diagram), and the unit is $[1]_{\mathcal{B}}$, the equivalence class of all linear combinations of Jacobi diagrams that evaluate to $\emptyset$ (see (2.4) for more on the unit).

Example 1.1.3. The IHX and AS relations are used to prove properties about Jacobi diagrams, by completing the diagrams of each term in a consistent manner. For example,

$$\begin{array}{c} \text{Loop with bottom edge marked} \end{array} = - \begin{array}{c} \text{Loop with top edge marked} \end{array} = - \begin{array}{c} \text{Loop with bottom edge marked} \end{array} \Rightarrow \begin{array}{c} \text{Loop with bottom edge marked} \end{array} \stackrel{\text{AS}}{\sim} 0$$

Graph isomorphism was applied in the last equality. To emphasize graph equality under the application of certain relations, the symbols $\stackrel{\text{AS}}{\sim}$, $\stackrel{\text{IHX}}{\sim}$ and $\stackrel{\mathcal{B}}{\sim}$ will be used, the latter meaning that either of IHX or AS have been applied in constructing the equalities.

Notation. Vertices of diagrams in $\mathcal{D}^o$ that are oriented in the default direction, counterclockwise, will be presented without a directional arrow, whereas vertices oriented clockwise will be marked by

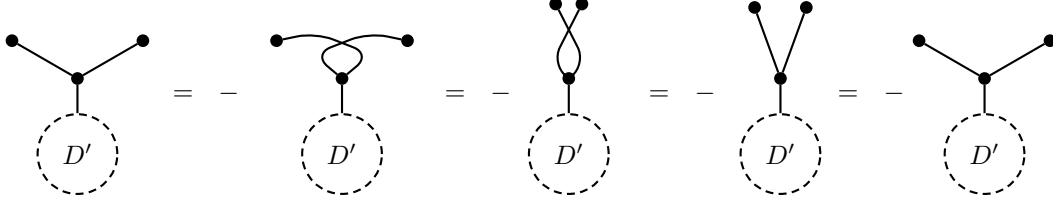


This allows us to restate the AS relation as

$$\text{AS: } \begin{array}{c} \text{Y-junction} \\ \text{with top edge marked} \end{array} = - \begin{array}{c} \text{Y-junction} \\ \text{with bottom edge marked and clockwise orientation} \end{array}$$

Lemma 1.1.4. If $D \in \mathcal{B}$ contains a trivalent vertex with two univalent neighbors, then $D \stackrel{\text{AS}}{\sim} 0$.

Proof: Suppose that D has a trivalent vertex t with two univalent neighbors t_1, t_2 . Denote by D' the subgraph of D that connects to t via the edge not ending in t_1 or t_2 . Apply the AS relation to t .

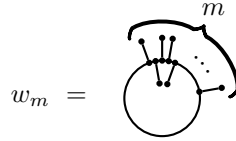


The result follows. ■

The lemma shows that every trivalent vertex has at most one univalent neighbor in every diagram in $\mathcal{B}$. Therefore every graph in $\mathcal{B}$ is a cubic graph with incident trees consisting of a single edge. This observation allows us to systematise the diagrams of $\mathcal{B}$.

Definition 1.1.5. (Ciliation) Given a graph G , if G' is obtained from G by attaching single edges to edges of G , in the process creating new trivalent vertices, then G' is termed a *ciliated* graph, obtained by *ciliating* G . The single edges are termed *cilia*. If m cilia are used in ciliating G , then G' is termed an *m-ciliation*.

Definition 1.1.6. (*m*-wheel) If G is an *m*-ciliation of a single loop, then G is termed an *m-wheel*. In this case, G is denoted by w_m .



The result of (1.1.4) may be generalized on the number of trivalent vertices of a given diagram.

Proposition 1.1.7. Given $D \in \mathcal{B}$, let $v_3(D)$ be the number of trivalent vertices of D . If D has an orientation-reversing automorphism, then $D = (-1)^{v_3(D)} D$.

Proof: Let f be an automorphism of D that reverses the orientation of every vertex of D . Apply the AS relation to every one of the trivalent vertices of D to get

$$D \stackrel{\text{AS}}{\sim} (-1)^{v_3(D)} f(D) = (-1)^{v_3(D)} D$$

This completes the proof. ■

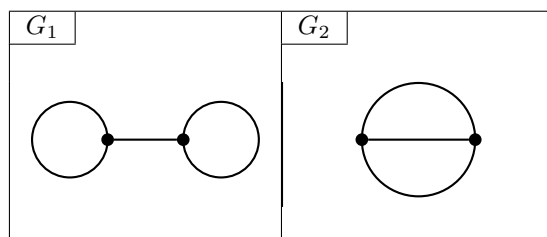
Note that if $v_3(D)$ is odd for $D \in \mathcal{B}$, then it directly follows that $D \stackrel{\mathcal{B}}{\sim} 0$. This shows the main use of the AS relation, to reduce diagrams to 0. As will be demonstrated in the following section, the IHX relation is used more to reduce diagrams to their constituent basis elements.

1.1.1 Connected elements of $\mathcal{B}$ of low degree

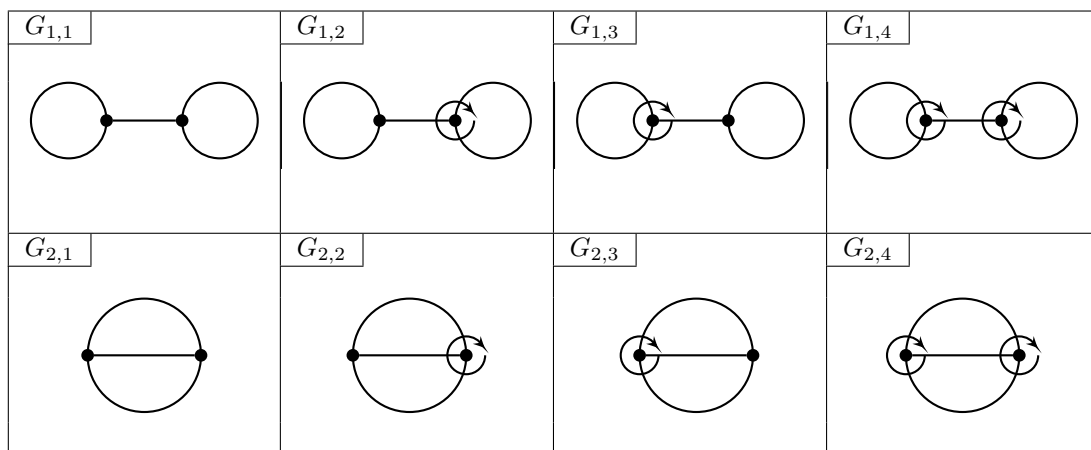
For D an open Jacobi diagram, $m = \deg(D) = \frac{1}{2}|V(D)|$, one-half the number of vertices of D . As $2|E| = \sum_{v \in V(D)} \deg(v) = 3|V(D)|$, it follows that $|E| = 3m$. We shall only consider cubic graphs with $2m$ vertices and $3m$ edges.

The space $\mathcal{D}_0^o$ of diagrams with no edges is empty.

Connected diagrams in $\mathcal{D}_1^o$ with 3 edges. The set of cubic graphs with 2 vertices and 3 edges is:



Hence in $\mathcal{D}_1^o$, there are at most $2 \cdot (2^2) = 8$ diagrams, by applying an ordering to the incident edges of each vertex.



We may identify pairs by smooth maps:

$$G_{1,1} = \text{---} \circ \text{---} \circ \text{---} \circ \text{---} = \text{---} \circ \text{---} \circ \text{---} \circ \text{---} = G_{1,4}$$

$$G_{1,2} = \text{---} \circ \text{---} \circ \text{---} \circ \text{---} = \text{---} \circ \text{---} \circ \text{---} \circ \text{---} = G_{1,3}$$

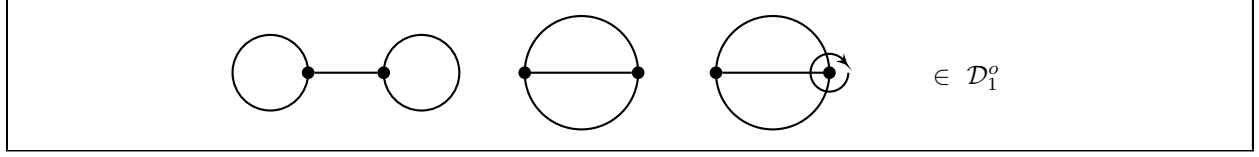
$$G_{2,1} = \text{---} \circ \text{---} \circ \text{---} \circ \text{---} = G_{2,4}$$

$$G_{2,2} = \text{---} \circ \text{---} \circ \text{---} \circ \text{---} = G_{2,3}$$

Each equality follows after a rotation through 180° around the indicated axis. This leaves 4 diagrams $G_{1,1}, G_{1,2}, G_{2,1}, G_{2,2}$. Next note that by a reversal of an orientation,

$$G_{1,2} = \text{diagram} = \text{diagram} = \text{diagram} = \text{diagram} = G_{1,1}$$

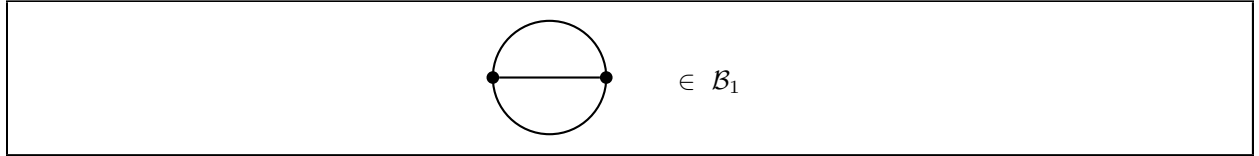
It follows that there are three distinct open Jacobi diagrams in $\mathcal{D}_1^o$:



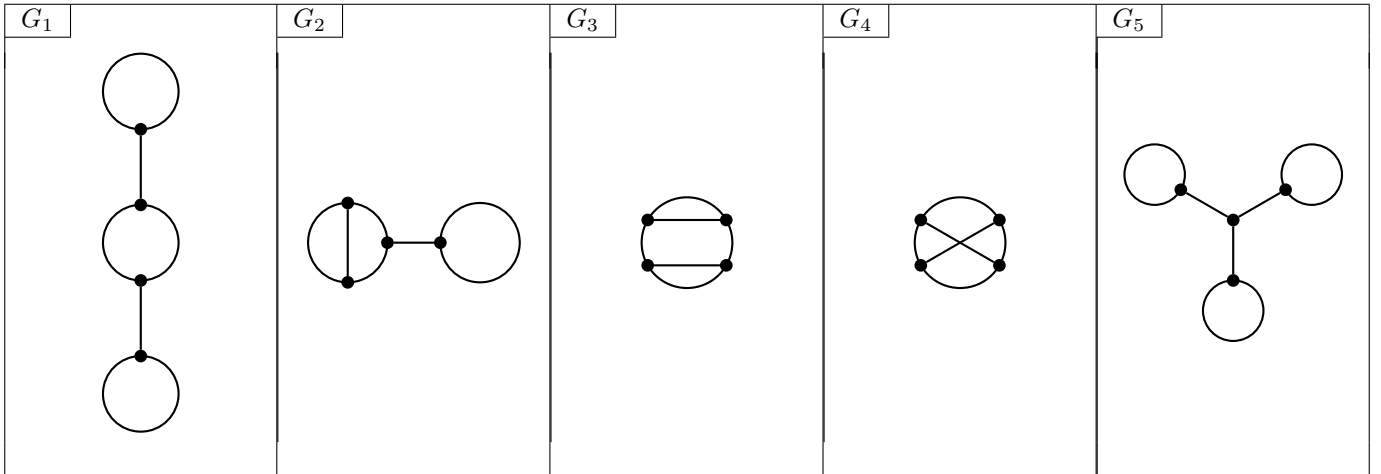
To see how they embed in $\mathcal{B}_1$, apply the AS and IHX relations. The observation from (1.1.3) of the AS relation indicates that $G_{1,1} \stackrel{\mathcal{B}}{\sim} 0$. By the IHX relation,

$$\text{diagram} = \text{diagram} - \text{diagram} = 0 - \text{diagram} = - \text{diagram} = - G_{2,2}$$

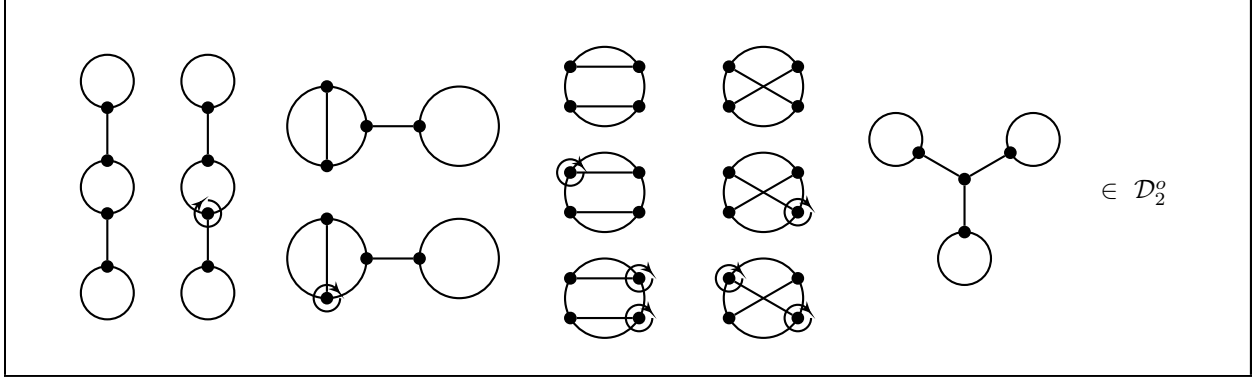
Hence $G_{2,1} \stackrel{\text{IHX}}{\sim} G_{2,2}$, and $\mathcal{D}_1^o$ contributes to $\mathcal{B}_1$ only the following diagram:



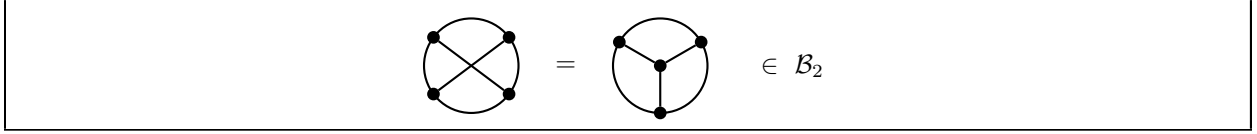
Connected diagrams in $\mathcal{D}_2^o$ with 6 edges. The set of cubic graphs with 4 vertices and 6 edges is:



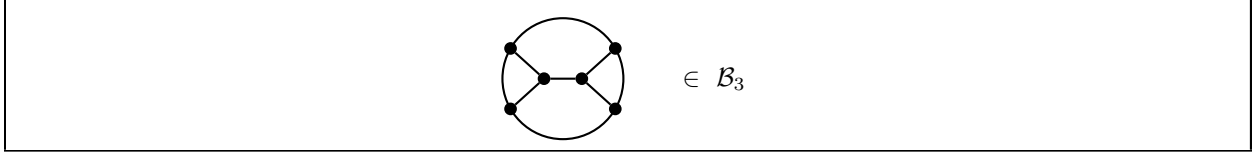
In $\mathcal{D}_2^o$ we have at most $5 \cdot (2^4) = 80$ diagrams. By calculations as above (see Appendix A), there are 11 distinct open Jacobi diagrams in $\mathcal{D}_2^o$.



Checking for independence modulo the IHX and AS relations, (see Appendix A), $\mathcal{B}_2$ has only 1 distinct diagram.



Connected diagrams in $\mathcal{D}_3^o$ with 9 edges. The number of cubic graphs with 6 vertices and 9 edges is 17, hence there are $17 \cdot 2^6 = 1088$ vertex-oriented cubic graphs with 6 vertices and 9 edges, leaving a single distinct element in $\mathcal{B}_3$:



The diagrams are reduced fully in Appendix B.

1.1.2 An alternative grading and bi-grading

Given a graph G , it is possible to obtain another bi-grading of the algebra. The rank of the fundamental group of a graph G gives a homotopy equivalence of graphs². The fundamental group is computed by contracting a spanning tree of each connected component and counting the loops emanating from the single vertex.

Definition 1.1.8. (loop-degree) Let G be a planar graph. Define the loop-degree of a graph G to be the rank of the fundamental group of G , or

$$\text{loop}(G) = \text{rank}(\pi_1(G))$$

Definition 1.1.9. (The vector space $\mathcal{D}_{n-\text{loop}}^o$) Define $\mathcal{D}_{n-\text{loop}}^o$ to be the freely-generated subspace of $\mathcal{D}^o$ of unitrivalent open Jacobi diagrams with loop-degree n .

²See [6], p.30

Proposition 1.1.10. The IHX and AS relations preserve loop-degree.

Proof: Since the graphs of the AS relation

$$\begin{array}{c} \diagup \\ \bullet \\ \diagdown \end{array} = - \begin{array}{c} \diagup \\ \bullet \\ \diagdown \end{array}$$

are isomorphic, loop-degree does not change. For the IHX relation, let e be the edge with both ends in each of the diagrams of the relation. By contracting e (which is not a loop, hence contraction does not affect loop-degree), it follows that

$$\begin{aligned} \text{loop} \left(\begin{array}{c} \text{---} \bullet \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \right) &= \text{loop} \left(\begin{array}{c} \diagup \bullet \diagdown \\ \diagdown \bullet \diagup \end{array} \right) \\ \text{loop} \left(\begin{array}{c} | \bullet | \\ \text{---} \bullet \text{---} \\ | \bullet | \end{array} \right) &= \text{loop} \left(\begin{array}{c} \diagup \bullet \diagdown \\ \diagdown \bullet \diagup \end{array} \right) \\ \text{loop} \left(\begin{array}{c} \diagup \bullet \diagdown \\ \diagdown \bullet \diagup \end{array} \right) &= \text{loop} \left(\begin{array}{c} \diagup \bullet \diagdown \\ \diagdown \bullet \diagup \end{array} \right) \end{aligned}$$

Hence the IHX relation does not affect loop-degree. ■

Using the fact that application of IHX and AS keeps the diagrams in a fixed space, we may formalize the grading.

Definition 1.1.11. (The vector spaces $\mathcal{B}_{n-\text{loop}}$ and $\mathcal{B}$) Define the vector spaces $\mathcal{B}$ and $\mathcal{B}_{n-\text{loop}}$ as

$$\mathcal{B}_{n-\text{loop}} = \frac{\mathcal{D}_{n-\text{loop}}^o}{(\text{IHX}, \text{AS})} \quad \mathcal{B} = \bigoplus_{n \geq 0} \mathcal{B}_{n-\text{loop}}$$

This gives a bi-grading of $\mathcal{B}$. Define $\mathcal{B}_{n,m}$ to be the subspace of $\mathcal{B}_{n-\text{loop}}$ of degree m diagrams, or equivalently, the subspace of $\mathcal{B}_m$ of diagrams with loop-degree n .

$$\mathcal{B} = \bigoplus_{n,m \geq 0} \mathcal{B}_{n,m}$$

1.2 The vector spaces $\mathcal{B}_{n,m}$

The basis of $\mathcal{B}_{n,m}$ is given by trivalent graphs found in Section 1.1.1 with total degree $\leq m$ and a single edge, which will be the cilia attached to the edges of the connected diagrams.

The vector space $\mathcal{B}_{0,m}$

If $m = 0$, there are no vertices, and hence no graphs. Therefore the span is $\emptyset$.

If $m \geq 1$, there are no loops, and the graph is a tree. Hence the span is a single edge

$$\left\{ \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \right\}$$

The vector space $\mathcal{B}_{1,m}$

If $m = 0$, there are no vertices, and hence no graphs. Thus the span is $\emptyset$.

If $m = 1$, there is one edge attached to a loop. Since the edge may be attached in two different ways, there are two graphs in this space:

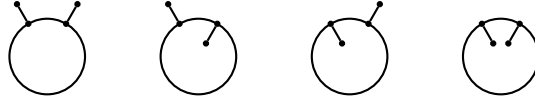
$$\left\{ \begin{array}{c} \text{---} \\ | \\ \bigcirc \end{array} , \begin{array}{c} \text{---} \\ \bigcirc \end{array} \right\}$$

However, by rotating the first graph about its center axis and reorienting the trivalent edge, we get the second graph. And a rotation followed by an application of AS gives the original graph back with a negative sign.

$$\begin{array}{c} \begin{array}{c} \text{---} \\ | \\ \bigcirc \end{array} \stackrel{\text{rot.}}{=} \begin{array}{c} \text{---} \\ \bigcirc \end{array} = \begin{array}{c} \text{---} \\ \bigcirc \end{array} \\ \begin{array}{c} \text{---} \\ | \\ \bigcirc \end{array} \stackrel{\text{rot.}}{=} \begin{array}{c} \text{---} \\ \bigcirc \end{array} \stackrel{\text{AS}}{=} - \begin{array}{c} \text{---} \\ | \\ \bigcirc \end{array} \end{array}$$

Hence the basis for $\mathcal{B}_{1,1}$ is null.

For $\mathcal{B}_{1,2}$, we have 4 different diagrams.



Applying a similar approach of rotation around the center axis, reorientation, and AS, we find that the space is the span of a single diagram.

$$\begin{array}{c} \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \stackrel{\text{rot.}}{=} \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} = \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \\ \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \stackrel{\text{AS}}{=} - \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \\ \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \stackrel{\text{AS}}{=} - \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \end{array}$$

Therefore the space $\mathcal{B}_{1,2}$ is

$$\left\{ \begin{array}{c} \text{---} \text{---} \\ \bigcirc \end{array} \right\}$$

As m increases, the patterns above generalize by the following proposition.

Proposition 1.2.1. When m is odd, $\mathcal{B}_{1,m}$ is empty, and when m is even, the basis of $\mathcal{B}_{1,m}$ contains only the m -wheel (as defined in (1.1.6)) with all cilia oriented in the same direction.

Proof: For m odd, let $D \in \mathcal{B}_{1,m}$. Fix the natural ordering $(c_1, \dots, c_m)$ of the cilia of D . Rotate the m -wheel about the central vertical axis.

$$D = \text{Diagram 1} \stackrel{\text{rot.}}{=} \text{Diagram 2}$$

Let $k = (m-1)/2$. For each $i = 1, 2, \dots, k$, if the current orientation of c_i is not the same as the original orientation of c_{m-i} , apply the AS relation to the trivalent vertices of cilia c_i and c_{m-i} .

Then the current orientation of all c_i for $i \neq k+1$ will be the same as the original orientation of c_{m-i} , the AS relation will have been applied an even number of times, and the ordering $(c_1, \dots, c_m)$ will have been reversed. Apply AS once to reorient c_{k+1} , whose orientation will necessarily have been reversed³. The result is the original diagram, with an odd number of AS applications, hence with a coefficient of -1 .

$$\text{Diagram 3} \stackrel{\text{AS}}{\sim} \text{Diagram 4} = -D$$

Hence $D \stackrel{\mathcal{B}}{\sim} 0$, and the space $\mathcal{B}_{1,m}$ is empty.

Now suppose that m is even. For $D \in \mathcal{B}_{2,m}$, there is $k \in [0, m]$ such that by k applications of AS, once to each trivalent vertex of the cilia not oriented in the desired direction, we get

$$D = \text{Diagram 5} \stackrel{\text{AS}}{\sim} (-1)^k \text{Diagram 6} = (-1)^k w_m$$

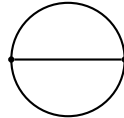
Hence either $D \stackrel{\mathcal{B}}{\sim} -w_m$ or $D \stackrel{\mathcal{B}}{\sim} w_m$, and $\mathcal{B}_{1,m} = \{w_m\}$, with all cilia oriented in a common direction. ■

It follows that for $n \in \mathbb{N}$, $\dim(\mathcal{B}_{1,2n}) = 1$ and $\dim(\mathcal{B}_{1,2n-1}) = 0$, and $\mathcal{B}_{1-\text{loop}}$ is given by

$$\mathcal{B}_{1-\text{loop}} = \text{span}_{\mathbb{C}}(\{\omega_a : a \in \mathbb{N}\})$$

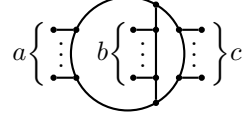
The vector space $\mathcal{B}_{2,m}$

In this vector space there is one unique connected diagram with 2 loops and no cilia (see Section 1.1.1), namely



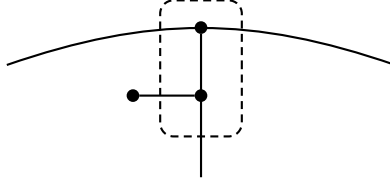
By using similar argumentation as in (1.2.1), we consider cilia of only one orientation (with respect to each edge) added to the diagram. Then for $a, b, c \in \mathbb{N}$ indicating the number of cilia on each edge, with $a + b + c = m-1$, the connected diagrams in $\mathcal{B}_{2,m}$ will look like

³The rotation may be viewed as a rotation about c_{k+1} , the middle cilium

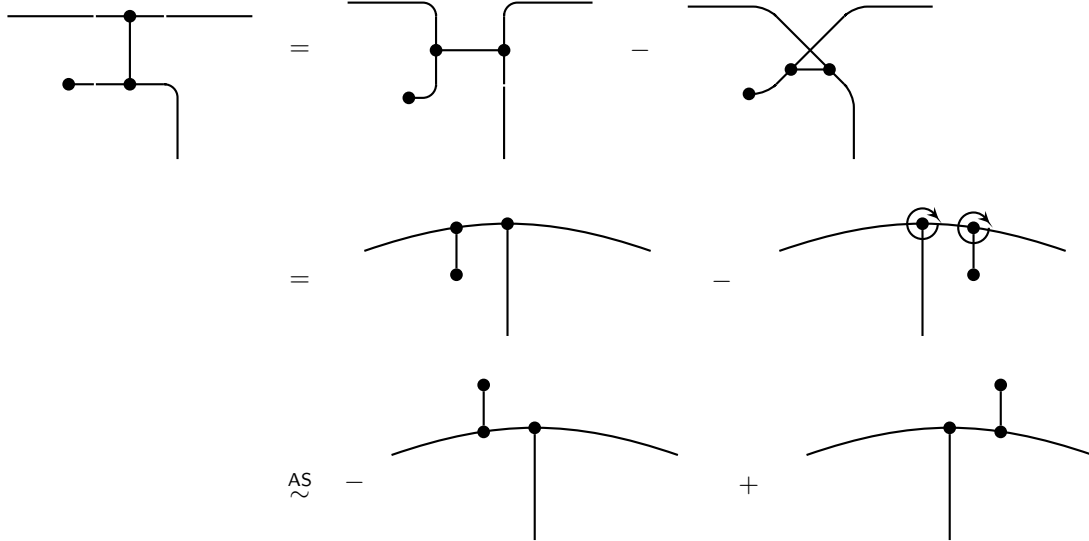


The disconnected diagrams of $\mathcal{B}_{2,m}$ will come from lower-dimensional spaces. With the IHX relation, it is possible to have $b = 0$ and group all the cilia on the two remaining edges.

Let (a, b, c) denote the diagram above, with the variables indicating the number of cilia on each edge. Consider the given portion of a diagram in $\mathcal{B}_{2,m}$, where the cilium is on the middle edge of the graph.



The selected section will be the 'T' in the IHX relation below.



Therefore $(a, b, c) = (a, b - 1, c + 1) - (a + 1, b - 1, c)$. This allows us to move all the cilia from the middle edge to the two outer edges by $2^b - 1$ applications of IHX, resulting in the relation

$$a \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} b \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} c = L \left(d \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} e \right)$$

where L is a linear map of diagrams, and $d + e = a + b + c = m - 1$ in each diagram on the right hand side. Hence $\mathcal{B}_{2,m}$ has connected diagrams with cilia on the two outer edges. The space $\mathcal{B}_{2-\text{loop}}$ is then given completely by

$$\mathcal{B}_{2-\text{loop}} = \text{span}_{\mathbb{C}} \left(\left\{ a \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} b : a, b \in \mathbb{N} \right\} \cup \left\{ \begin{array}{c} c \\ \vdots \\ \vdots \end{array} \right\} \begin{array}{c} d \\ \vdots \\ \vdots \end{array} : c, d \in \mathbb{N} \right)$$

Note that disconnected diagrams also have to be taken into account.

The spaces $\mathcal{B}_{2,m}$ in $\mathcal{B}_{2\text{-loop}}$ are not all the same, however. Recall that $a + b = m - 1$ and $c + d = m$ in $\mathcal{B}_{2,m}$. When n is even, the diagrams of the first type all vanish, as rotating about the horizontal axis,

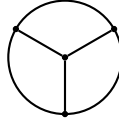
$$a \left\{ \begin{array}{c} \vdots \\ \text{---} \end{array} \right\} \left(\text{circle with vertical line} \right) \left\{ \begin{array}{c} \vdots \\ \text{---} \end{array} \right\} b \stackrel{\text{rot.}}{=} a \left\{ \begin{array}{c} \vdots \\ \text{---} \end{array} \right\} \left(\text{circle with vertical line and four trivalent vertices} \right) \left\{ \begin{array}{c} \vdots \\ \text{---} \end{array} \right\} b \stackrel{\text{AS}}{\approx} (-1)^{n+1} a \left\{ \begin{array}{c} \vdots \\ \text{---} \end{array} \right\} \left(\text{circle with vertical line} \right) \left\{ \begin{array}{c} \vdots \\ \text{---} \end{array} \right\} b$$

Moreover, for m even, only diagrams of the second type with c and d both even are non-zero. Hence for $n \in \mathbb{N}$, $\dim(\mathcal{B}_{2,2n}) = \lfloor n/2 \rfloor$, which follows from the number of ways to split $2n$ into two groups of even size.

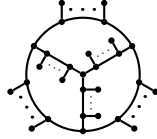
When m is odd, every diagram of the second type evaluates to zero. So for $n \in \mathbb{N}$, $\dim(\mathcal{B}_{2,2n-1}) = n$, which follows from the number of distinct ways to split a set of size $2n - 1$ in two.

The vector space $\mathcal{B}_{3,m}$

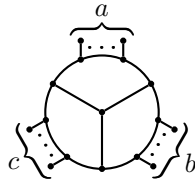
In this vector space there is one unique connected diagram with 3 loops and no cilia, namely



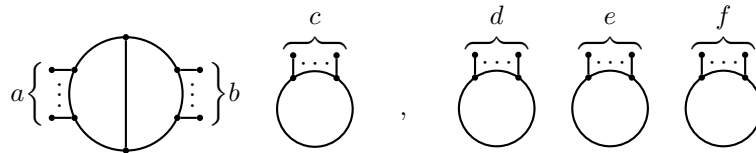
Cilia may be placed on each edge, so diagrams in $\mathcal{B}_{3,m}$ look like



Apply the same procedure as in the previous example to move cilia from the inside edges to the outer edges. Then $\mathcal{B}_{3,m}$ contains diagrams of the type



where $a + b + c = m - 2$. The space also has disconnected diagrams, namely



for which $a + b + c = m - 1$ and $d + e + f = m$. The dimension of $\mathcal{B}_{3,m}$ is at most the number of partitions of $m - 2$, $m - 1$, and m into 3-element sets. The size is limited in several ways, for each type of diagram:

The connected diagram evaluates to zero when $a + b + c$ is odd (m is odd), and a, b, c are not all distinct. Then a rotation through any axis and application of the AS relation once to each cilium's trivalent vertex, for a total of m applications, will give the original diagram back, with a coefficient of -1 .

The diagram with two connected components evaluates to zero when m is even, as then either the 2-loop or the 1-loop component has an odd number of cilia, and in both cases that will give $D = -D$ for D this diagram. When m is odd, if both have an odd number of cilia, then the diagram evaluates to zero.

The diagram with 3 connected components evaluates to zero when m is odd, as dividing an odd number into three parts means at least one part has an odd number of cilia. When m is even, the diagram evaluates to zero only if m is not divided into three even parts.

The space $\mathcal{B}_{3\text{-loop}}$ is then given by

$$\mathcal{B}_{3\text{-loop}} = \text{span}_{\mathbb{C}} \left(\left\{ \begin{array}{c} \text{Diagram 1} \\ : a, b, c \in \mathbb{N} \end{array} \right\} \cup \left\{ \begin{array}{c} \text{Diagram 2} \\ : d, e, f \in \mathbb{N} \end{array} \right\} \cup \left\{ \begin{array}{c} \text{Diagram 3} \\ : g, h, i \in \mathbb{N} \end{array} \right\} \right)$$

The succeeding sections deal with the original grading of $\mathcal{B}$ by degree rather than by loop-degree.

2 The Hopf algebras induced by $\mathcal{B}$ and $\mathcal{A}$

A Hopf algebra is a bi-algebra that is a unital associative algebra and a co-unital co-associative co-algebra. The goal of this section is to construct such a bi-algebra $\mathfrak{B} = (\mathcal{B}, \sqcup, \overline{\Delta}, \overline{\eta}, \overline{\varepsilon})$ over a space $\mathcal{B}$ with a product $\sqcup$, a co-product $\overline{\Delta}$, a unit $\overline{\eta}$ and a co-unit $\overline{\varepsilon}$.

The Hopf algebra structure of $\mathcal{A}$ will also be discussed, although only as far as it is necessary to construct the desired isomorphism. For a more detailed account of $\mathcal{A}$, refer to [3].

The link between $\mathfrak{A}$ and $\mathfrak{B}$ comes from the fact that the universal enveloping algebra of a semi-simple algebraic group is the dual of the Hopf algebra [1]. Note that in the process of constructing these algebras, often we refer to the algebra itself and the vector space of its elements by the same symbol. Where ambiguous, it will be made clear what object is being discussed, but most often it will be clear from the context.

2.1 $\mathcal{D}$ and $\mathcal{B}$ as bi-algebras

Lemma 2.1.1. $\mathcal{D} \doteq (\mathcal{D}^o, \sqcup, \Delta, \eta, \varepsilon)$ is an associative, commutative, co-associative, co-commutative bi-algebra with unit and co-unit, such that for all $k_1, k_2 \in \mathbb{C}$ and $D, D_1, D_2 \in \mathcal{D}^o$,

· product

$$\begin{aligned} \sqcup : \quad \mathcal{D}^o \otimes \mathcal{D}^o &\rightarrow \mathcal{D}^o \\ (k_1 D_1) \otimes (k_2 D_2) &\mapsto (k_1 k_2)(D_1 \sqcup D_2) \end{aligned}$$

· co-product

$$\begin{aligned} \Delta : \quad \mathcal{D}^o &\rightarrow \mathcal{D}^o \otimes \mathcal{D}^o \\ D &\mapsto \sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \end{aligned}$$

· unit

$$\begin{aligned} \eta : \quad \mathbb{C} &\rightarrow \mathcal{D}^o \\ 1_{\mathbb{C}} &\mapsto 1_{\mathcal{D}^o} = 1 \cdot \emptyset \end{aligned}$$

· co-unit

$$\begin{aligned} \varepsilon : \quad \mathcal{D}^o &\rightarrow \mathbb{C} \\ D &\mapsto \begin{cases} 1_{\mathbb{C}} & \text{if } D = 1_{\mathcal{D}^o} \\ 0_{\mathbb{C}} & \text{else} \end{cases} \end{aligned}$$

Proof: The linearity of the product $\sqcup$ follows from the linearity of $\otimes$. The commutativity and associativity of the product comes from the commutativity and associativity of disjoint union, so the product is well-defined. As the tensor product is well defined, the co-product is well-defined, but co-commutativity and co-associativity must be checked.

· *Co-commutativity of the co-product:* It must be shown that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{D}^o \otimes \mathcal{D}^o & \xleftarrow{\tau} & \mathcal{D}^o \otimes \mathcal{D}^o \\ & \Delta \swarrow \quad \searrow \Delta & \\ & \mathcal{D}^o & \end{array}$$

where $\tau : \mathcal{D}^o \otimes \mathcal{D}^o \rightarrow \mathcal{D}^o \otimes \mathcal{D}^o$ is the twist isomorphism, given by $\tau : D_1 \otimes D_2 \mapsto D_2 \otimes D_1$. The above diagram commutes if and only if $\tau \circ \Delta = \Delta$. Consider $D \in \mathcal{D}^o$, and as $\otimes$ is a commutative product⁴,

$$\Delta(D) = \sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 = \sum_{D_2 \sqcup D_1 = D} D_2 \otimes D_1 = \tau(\Delta(D)) = (\tau \circ \Delta)(D)$$

This proves the diagram commutes, so the bi-algebra is co-commutative.

· *Co-associativity of the co-product:* It must be shown that $(\Delta \otimes \text{id}_{\mathcal{D}^o}) \circ \Delta = (\text{id}_{\mathcal{D}^o} \otimes \Delta) \circ \Delta$. The identity map on $\mathcal{D}^o$ is multiplication by $1_{\mathbb{C}} \cdot \emptyset$, so

$$\begin{aligned} (\Delta \otimes 1)(\Delta(D)) &= (\Delta \otimes 1) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) & (1 \otimes \Delta)(\Delta(D)) &= (1 \otimes \Delta) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) \\ &= \sum_{D_1 \sqcup D_2 = D} \left(\sum_{D_3 \sqcup D_4 = D_1} D_3 \otimes D_4 \right) \otimes D_2 & &= \sum_{D_1 \sqcup D_2 = D} D_1 \otimes \left(\sum_{D_3 \sqcup D_4 = D_2} D_3 \otimes D_4 \right) \\ &= \sum_{D_3 \sqcup D_4 \sqcup D_2 = D} (D_3 \otimes D_4) \otimes D_2 & &= \sum_{D_1 \sqcup D_3 \sqcup D_4 = D} D_1 \otimes (D_3 \otimes D_4) \\ &= \sum_{D_1 \sqcup D_2 \sqcup D_3 = D} D_1 \otimes D_2 \otimes D_3 & &= \sum_{D_1 \sqcup D_2 \sqcup D_3 = D} D_1 \otimes D_2 \otimes D_3 \end{aligned}$$

· *Unit:* It is clear that η is a unit.

· *Co-unit:* To show that the co-unit ε is actually a co-unit, we first define two functions

$$\begin{array}{ccc} \iota : \mathcal{D}^o & \rightarrow & \mathbb{C} \otimes \mathcal{D}^o \\ D & \mapsto & 1_{\mathbb{C}} \otimes D \end{array} \quad \begin{array}{ccc} \iota' : \mathcal{D}^o & \rightarrow & \mathcal{D}^o \otimes \mathbb{C} \\ D & \mapsto & D \otimes 1_{\mathbb{C}} \end{array} \quad (2.1)$$

With these functions, it is necessary to show that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{D}^o \otimes \mathcal{D}^o & \xrightarrow{\varepsilon \otimes \text{id}} & \mathbb{C} \otimes \mathcal{D}^o \\ & \Delta \swarrow \quad \searrow \iota & \\ & \mathcal{D}^o & \\ & \Delta \swarrow \quad \searrow \iota' & \\ \mathcal{D}^o \otimes \mathcal{D}^o & \xrightarrow{\text{id} \otimes \varepsilon} & \mathcal{D}^o \otimes \mathbb{C} \end{array} \quad (2.2)$$

⁴see [3] Ch. 12 and [4] Sec. 2

Now we show that $(\varepsilon \otimes \text{id}) \circ \Delta = \iota$ and $(\text{id} \otimes \varepsilon) \circ \Delta = \iota'$. First consider the upper part of the diagram,

$$(\varepsilon \otimes \text{id}) \circ \Delta(D) = (\varepsilon \otimes \text{id}) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) = \sum_{D_1 \sqcup D_2 = D} \varepsilon(D_1) \otimes D_2 = 1_{\mathbb{C}} \otimes D$$

The last equality follows as every term where $D_2 \neq D$ has $D_1 \neq \emptyset$, so then $\varepsilon(D_1) = 0_{\mathbb{C}}$. For the lower part of the diagram,

$$(\text{id} \otimes \varepsilon) \circ \Delta(D) = (\text{id} \otimes \varepsilon) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) = \sum_{D_1 \sqcup D_2 = D} D_1 \otimes \varepsilon(D_2) = D \otimes 1_{\mathbb{C}}$$

with consequences exactly as for the upper part. Therefore the diagram commutes, and the co-unit is indeed a co-unit. This completes the proof. $\blacksquare$

Having induced a bi-algebra structure in $\mathcal{D}$, we now proceed to do the same for $\mathcal{B}$. As $\mathcal{B}$ is a quotient space (from (1.1)), we will denote elements in $\mathcal{B}$ by their class representatives $[D]_{\mathcal{B}}$, where $D_1 \in [D_2]_{\mathcal{B}}$ if and only if $D_1 \stackrel{\mathcal{B}}{\sim} D_2$.

Lemma 2.1.2. $\mathcal{B} = (\mathcal{B}, \sqcup, \overline{\Delta}, \overline{\eta}, \overline{\varepsilon})$ is an associative, commutative, co-associative, co-commutative bi-algebra with unit and co-unit, such that for all $D, D_1, D_2 \in \mathcal{B}$,

· product

$$\begin{aligned} \sqcup : \quad \mathcal{B} \otimes \mathcal{B} &\rightarrow \mathcal{B} \\ [D_1]_{\mathcal{B}} \otimes [D_2]_{\mathcal{B}} &\mapsto [D_1 \sqcup D_2]_{\mathcal{B}} \end{aligned}$$

· co-product

$$\begin{aligned} \overline{\Delta} : \quad \mathcal{B} &\rightarrow \mathcal{B} \otimes \mathcal{B} \\ [D]_{\mathcal{B}} &\mapsto \sum_{D_1 \sqcup D_2 = D} [D_1]_{\mathcal{B}} \otimes [D_2]_{\mathcal{B}} \end{aligned}$$

· unit

$$\begin{aligned} \overline{\eta} : \quad \mathbb{C} &\rightarrow \mathcal{B} \\ 1_{\mathbb{C}} &\mapsto [1]_{\mathcal{B}} \end{aligned}$$

· co-unit

$$\begin{aligned} \overline{\varepsilon} : \quad \mathcal{B} &\rightarrow \mathbb{C} \\ [D]_{\mathcal{B}} &\mapsto \begin{cases} 1_{\mathbb{C}} & \text{if } [D]_{\mathcal{B}} = [1]_{\mathcal{B}} \\ 0_{\mathbb{C}} & \text{else} \end{cases} \end{aligned}$$

Proof: · *Product:* To show that the product $\sqcup$ is well defined, consider $D'_1 \in [D_1]_{\mathcal{B}}$ and $D'_2 \in [D_2]_{\mathcal{B}}$. Then $D_1 \stackrel{\mathcal{B}}{\sim} D'_1$ and $D_2 \stackrel{\mathcal{B}}{\sim} D'_2$. As the product operator takes the disjoint union of open Jacobi diagrams, the AS and IHX relations may be applied to each component separately, yielding

$$D_1 \sqcup D_2 \stackrel{\mathcal{B}}{\sim} D'_1 \sqcup D_2 \stackrel{\mathcal{B}}{\sim} D'_1 \sqcup D'_2$$

Thus $\sqcup$ is well-defined.

· *Co-product:* To show that the co-product $\overline{\Delta}$ is well defined, it must be shown that it is independent of the representative element D . So let $D \stackrel{\mathcal{B}}{\sim} D'$, so each connected component of D is equal, modulo the IHX and AS relations, to a connected component of D' . That is,

$$\begin{aligned} D &= D_1 \sqcup D_2 \sqcup \dots \sqcup D_k \\ D' &= D'_1 \sqcup D'_2 \sqcup \dots \sqcup D'_k \\ \implies D_1 &\stackrel{\mathcal{B}}{\sim} D'_1, D_2 \stackrel{\mathcal{B}}{\sim} D'_2, \dots, D_k \stackrel{\mathcal{B}}{\sim} D'_k \end{aligned}$$

Consider the action of the product on D . Let $\{1, 2, \dots, k\} = I_1 \sqcup I_2$ for I_1, I_2 disjoint and nonempty. Then $\bigsqcup_{i \in I_1} D_i \stackrel{\mathcal{B}}{\sim} \bigsqcup_{i \in I_1} D'_i$ for fixed I_1 , by working element-wise on the expressions. Hence

$$\begin{aligned} \Delta(D) &= (1_{\mathcal{B}} \otimes D) + (D \otimes 1_{\mathcal{B}}) + \sum_{I_1, I_2} \left(\bigsqcup_{i \in I_1} D_i \otimes \bigsqcup_{j \in I_2} D_j \right) \\ &\stackrel{\mathcal{B}}{\sim} (1_{\mathcal{B}} \otimes D') + (D' \otimes 1_{\mathcal{B}}) + \sum_{I_1, I_2} \left(\bigsqcup_{i \in I_1} D'_i \otimes \bigsqcup_{j \in I_2} D'_j \right) \\ &= \Delta(D') \end{aligned}$$

Let $\pi : D^o \rightarrow \mathcal{B}$ be the natural projection, so we have $\overline{\Delta}([D]_{\mathcal{B}}) = (\pi \otimes \pi)(\Delta(D))$, and applying the above,

$$\overline{\Delta}([D]_{\mathcal{B}}) = (\pi \otimes \pi)(\Delta(D)) = (\pi \otimes \pi)(\Delta(D')) = \overline{\Delta}([D']_{\mathcal{B}}) \quad (2.3)$$

This shows that $\overline{\Delta}$ is independent of the class representative, and so is well-defined.

- *Commutativity and associativity of product:* Follows from (2.1.1) as $\sqcup$ and $\sqcap$ operate component-wise.
- *Co-commutativity of the co-product:* Follows from (2.1.1) as $\otimes$ is commutative for elements and element equivalence classes.
- *Co-associativity of the co-product:* We must prove that

$$(\text{id} \otimes \overline{\Delta}) \circ \overline{\Delta}([D]_{\mathcal{B}}) = (\overline{\Delta} \otimes \text{id}) \circ \overline{\Delta}([D]_{\mathcal{B}})$$

Using the above calculations, we have that

$$\begin{aligned} ((\text{id} \otimes \overline{\Delta}) \circ \overline{\Delta})([D]_{\mathcal{B}}) &= (\text{id} \otimes \overline{\Delta})(\overline{\Delta}([D]_{\mathcal{B}})) \\ &= (\text{id} \otimes \overline{\Delta})((\pi \otimes \pi)(\Delta(D))) \\ &= (\pi \otimes (\overline{\Delta} \circ \pi))(\Delta(D)) \\ &= (\pi \otimes (\overline{\Delta} \circ \pi)) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) \\ &= \sum_{D_1 \sqcup D_2 = D} \pi(D_1) \otimes (\overline{\Delta} \circ \pi)(D_2) \\ &= \sum_{D_1 \sqcup D_2 = D} \pi(D_1) \otimes \overline{\Delta}([D_2]_{\mathcal{B}}) \\ &= \sum_{D_1 \sqcup D_2 = D} \pi(D_1) \otimes ((\pi \otimes \pi) \circ \Delta)(D_2) && \text{(from (2.3))} \\ &= (\pi \otimes ((\pi \otimes \pi) \circ \Delta))(\Delta(D)) \\ &= ((\pi \otimes \pi \otimes \pi) \circ (\text{id} \otimes \Delta))(\Delta(D)) \\ &= ((\pi \otimes \pi \otimes \pi) \circ (\Delta \otimes \text{id}))(\Delta(D)) \\ &= (((\pi \otimes \pi) \circ \Delta) \otimes \pi)(\Delta(D)) \end{aligned}$$

$$\begin{aligned}
&= (((\pi \otimes \pi) \circ \Delta) \otimes \pi) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) \\
&= \sum_{D_1 \sqcup D_2 = D} (\pi \otimes \pi)(\Delta(D_1)) \otimes \pi(D_2) \quad (\text{from (2.3)}) \\
&= \sum_{D_1 \sqcup D_2 = D} \overline{\Delta}([D_1]_{\mathcal{B}}) \otimes [D_2]_{\mathcal{B}} \\
&= (\overline{\Delta} \otimes \text{id}) \circ \overline{\Delta}([D]_{\mathcal{B}})
\end{aligned}$$

This proves that $\overline{\Delta}$ is co-associative.

· *Unit*: It remains to show that the unit is independent of the choice of representative. Suppose that $D' \stackrel{\mathcal{B}}{\sim} [1]_{\mathcal{B}}$ and let $D \in \mathcal{B}$. As $\stackrel{\mathcal{B}}{\sim}$ operates component-wise,

$$[D']_{\mathcal{B}} \sqcup [D]_{\mathcal{B}} \stackrel{\mathcal{B}}{\sim} [1]_{\mathcal{B}} \sqcup [D]_{\mathcal{B}} = [D]_{\mathcal{B}} \quad (2.4)$$

Therefore the choice of representative is not important, and $[1]_{\mathcal{B}}$ is the unit in $\mathcal{B}$.

· *Co-unit*: A diagram analogous to (2.2) must be shown to commute. We employ the function symbols from (2.1) analogously (i.e. $[D]_{\mathcal{B}}$ instead of D , etc).

$$(\overline{\varepsilon} \otimes \text{id}) \circ \overline{\Delta}([D]_{\mathcal{B}}) = (\overline{\varepsilon} \otimes \text{id}) \left(\sum_{D_1 \sqcup D_2 = D} [D_1]_{\mathcal{B}} \otimes [D_2]_{\mathcal{B}} \right) = \sum_{D_1 \sqcup D_2 = D} \overline{\varepsilon}([D_1]_{\mathcal{B}}) \otimes [D_2]_{\mathcal{B}} = 1_{\mathbb{C}} \otimes [D]_{\mathcal{B}}$$

For the lower part of the diagram,

$$(\text{id} \otimes \overline{\varepsilon}) \circ \overline{\Delta}([D]_{\mathcal{B}}) = (\text{id} \otimes \overline{\varepsilon}) \left(\sum_{D_1 \sqcup D_2 = D} [D_1]_{\mathcal{B}} \otimes [D_2]_{\mathcal{B}} \right) = \sum_{D_1 \sqcup D_2 = D} [D_1]_{\mathcal{B}} \otimes \overline{\varepsilon}([D_2]_{\mathcal{B}}) = [D]_{\mathcal{B}} \otimes 1_{\mathbb{C}}$$

Therefore $\overline{\varepsilon}$ is a co-unit, and the lemma is proved. ■

2.2 The Hopf algebra $\mathfrak{B}$

Combining lemmata (2.1.1) and (2.1.2) to prove the following proposition, the main theorem (2.2.3) will follow.

Proposition 2.2.1. $\mathfrak{B} = (\mathcal{B}, \sqcup, \overline{\Delta}, \overline{\eta}, \overline{\varepsilon})$ is a commutative, co-commutative, connected finite-type graded bi-algebra with unit and co-unit, such that for all $D, D_1, D_2 \in \mathcal{B}$,

· product

$$\begin{aligned}
\sqcup : \quad \mathfrak{B} \otimes \mathfrak{B} &\rightarrow \mathfrak{B} \\
[D_1]_{\mathfrak{B}} \otimes [D_2]_{\mathfrak{B}} &\mapsto [D_1 \sqcup D_2]_{\mathfrak{B}}
\end{aligned}$$

· co-product

$$\begin{aligned}
\overline{\Delta} : \quad \mathfrak{B} &\rightarrow \mathfrak{B} \otimes \mathfrak{B} \\
[D]_{\mathfrak{B}} &\mapsto \sum_{D_1 \sqcup D_2 = D} [D_1]_{\mathfrak{B}} \otimes [D_2]_{\mathfrak{B}}
\end{aligned}$$

· unit

$$\begin{aligned}
\overline{\eta} : \quad \mathbb{C} &\rightarrow \mathfrak{B} \\
1_{\mathbb{C}} &\mapsto [1]_{\mathfrak{B}}
\end{aligned}$$

· co-unit

$$\begin{aligned} \bar{\varepsilon} : \quad \mathfrak{B} &\rightarrow \mathbb{C} \\ [D]_{\mathfrak{B}} &\mapsto \begin{cases} 1_{\mathbb{C}} & \text{if } [D]_{\mathfrak{B}} = [1]_{\mathfrak{B}} \\ 0_{\mathbb{C}} & \text{else} \end{cases} \end{aligned}$$

The grading is the one introduced in (1.1.2), by the degree of the diagrams, with $\mathfrak{B}_0$ the space containing only $[1]_{\mathfrak{B}}$, the identity diagram.

Proof: The product, co-product, and unit may be extended linearly from $\mathcal{B}$. It remains to check the other properties.

· *Co-unit:* For an element $[D]_{\mathfrak{B}}$ of $\mathfrak{B}$, $[D]_{\mathfrak{B}} = [1]_{\mathfrak{B}}$ if and only if $[D]_{\mathfrak{B}} \in \mathfrak{B}_0$. Hence it follows that, for $z \in \mathbb{C}$, $[D]_{\mathfrak{B}} = z[1]_{\mathfrak{B}}$ if and only if $\bar{\varepsilon}([D]_{\mathfrak{B}}) = z$. Therefore the described co-unit satisfies the properties for being a co-unit.

· *Co-commutativity of the co-product:* Follows from (2.1.2) as $\otimes$ is commutative for equivalence classes.

· *Grading as an algebra:* For $n, m \in \mathbb{N}$, let $[D_1]_{\mathfrak{B}} \in \mathfrak{B}_n$ and $[D_2]_{\mathfrak{B}} \in \mathfrak{B}_m$, so D_1 has degree n (i.e. $2n$ vertices) and D_2 has degree m (i.e. $2m$ vertices). Then

$$[D_1]_{\mathfrak{B}} \sqcup [D_2]_{\mathfrak{B}} = [D_1 \sqcup D_2]_{\mathfrak{B}} \in \mathfrak{B}_{n+m}$$

This follows as the disjoint union of D_1 and D_2 has $2n+2m = 2(n+m)$ vertices, and hence is of degree $n+m$.

· *Grading as a co-algebra:* For $n \in \mathbb{N}$, let $[D] \in \mathcal{B}_n$. Then

$$\bar{\Delta}([D]_{\mathfrak{B}}) = \sum_{D_1 \sqcup D_2 = D} [D_1]_{\mathfrak{B}} \otimes [D_2]_{\mathfrak{B}} \in \bigoplus_{a+b=n} \mathfrak{B}_a \otimes \mathfrak{B}_b$$

This follows as for every $[D_1]_{\mathfrak{B}}, [D_2]_{\mathfrak{B}} \in \mathfrak{B}$ with $D_1 \sqcup D_2 = D$ for $[D]_{\mathfrak{B}} \in \mathfrak{B}_n$, there exist some $a, b \in \mathbb{N}$ with $a+b=n$, so $[D_1]_{\mathfrak{B}} \in \mathfrak{B}_a$ and $[D_2]_{\mathfrak{B}} \in \mathfrak{B}_b$, and hence $[D_1]_{\mathfrak{B}} \otimes [D_2]_{\mathfrak{B}} = [D]_{\mathfrak{B}} \in \mathfrak{B}_n = \mathfrak{B}_a \otimes \mathfrak{B}_b$.

· *Finite type:* For $n \in \mathbb{N}$, $\mathfrak{B}_n$ has finitely many elements and hence is finitely generated.

· *Connected:* It is left to show that $\bar{\varepsilon}|_{\mathfrak{B}_0} : \mathfrak{B}_0 \rightarrow \mathbb{C}$ given by $z[1]_{\mathfrak{B}} \mapsto z$ is an isomorphism, for $z \in \mathbb{C}$. This follows immediately from the definition of the co-unit above. ■

To prove the final theorem of this section, it is necessary to use the Milnor-Moore theorem. For a complete proof, refer to [3].

Theorem 2.2.2. [MILNOR, MOORE]

Let $\mathfrak{A}$ be a commutative, co-commutative connected finite-type graded bi-algebra with unit and co-unit over a field of characteristic zero. Then:

- i. $\mathfrak{A}$ is isomorphic to the symmetric algebra of its primitives
- ii. The isomorphism in i. is a natural extension of the inclusion of the primitives of $\mathfrak{A}$ into $\mathfrak{A}$
- iii. $\mathfrak{A}$ has no zero divisors
- iv. $\mathfrak{A}$ is a Hopf algebra with an anti-homomorphism $h : p \mapsto -p$, for p in the symmetric algebra of $\mathfrak{A}$

■

Point iv. is of interest to us. The bi-algebra $\mathfrak{B}$ fulfills the preconditions of the Milnor-Moore theorem by (2.3.1), and as $\text{char}(\mathbb{C}) = 0$, the succeeding theorem and (2.3.3) follow immediately.

Theorem 2.2.3. $\mathfrak{B} = (\mathcal{B}, \sqcup, \bar{\Delta}, \bar{\eta}, \bar{\varepsilon})$ is a Hopf algebra.

■

We now briefly consider the Hopf algebra structure of $\mathfrak{A}$.

2.3 The Hopf algebra $\mathfrak{A}$

The product Δ has been defined on $\mathcal{D}^o$, but it can also be viewed as a map on $\mathcal{A}$. The following definition is more of an explanation of how Δ affects $\mathcal{A}$.

Definition 2.3.1. Every $D \in \mathcal{A}$ consists of a set Y of connected trivalent graphs and a circular skeleton. For every $X \subset Y$, let D_X denote D with the graphs in $Y - X$ deleted. Then the product on $\mathcal{A}$ is given by

$$\begin{aligned} \Delta : \mathcal{A} &\rightarrow \mathcal{A} \otimes \mathcal{A} \\ D &\mapsto \sum_{X \subset Y} D_X \otimes D_{Y-X} \end{aligned}$$

The following propositions are presented for completeness; no proofs will be given.

Proposition 2.3.2. $\mathcal{A} = (\mathcal{A}, \#, \overline{\Delta}, \overline{\eta}, \overline{\varepsilon})$ is a commutative, co-commutative, connected finite-type graded bi-algebra with unit and co-unit, such that for all $k, k_1, k_2 \in \mathbb{C}$ and $D, D_1, D_2 \in \mathcal{A}$,

· product

$$\begin{aligned} \# : \mathcal{A} \otimes \mathcal{A} &\rightarrow \mathcal{A} \\ (k_1 D_1) \otimes (k_2 D_2) &\mapsto (k_1 k_2) (D_1 \# D_2) \end{aligned}$$

· co-product

$$\begin{aligned} \overline{\Delta} : \mathcal{A} &\rightarrow \mathcal{A} \otimes \mathcal{A} \\ [D] &\mapsto \sum_{X \subset Y} [D_X] \otimes [D_{Y-X}] \end{aligned}$$

· unit

$$\overline{\eta} : k \rightarrow k \cdot \text{[circular skeleton with arrow]}$$

· co-unit

$$\overline{\varepsilon} : [D] \rightarrow \begin{cases} 1_{\mathbb{C}} & \text{if } D \approx^{\mathcal{A}} \text{[circular skeleton with arrow]} \\ 0_{\mathbb{C}} & \text{else} \end{cases}$$

■

Theorem 2.3.3. $\mathfrak{A} = (\mathcal{A}, \#, \overline{\Delta}, \overline{\eta}, \overline{\varepsilon})$ is a Hopf algebra.

■

With the Hopf algebra structure on $\mathfrak{A}$ and $\mathfrak{B}$, we continue with a description of the function that identifies them.

3 An isomorphism between $\mathfrak{A}$ and $\mathfrak{B}$

Note that both $\mathfrak{A}$ and $\mathfrak{B}$ were generated by diagrammizations of algebras, $\mathfrak{U}(\mathfrak{g})$ and $\mathfrak{S}(\mathfrak{g})$, respectively, for $\mathfrak{g}$ a metrized Lie algebra with certain properties. Their difference lies in the fact that $\mathfrak{B}$ is a Hopf algebra of open Jacobi diagrams, and $\mathfrak{A}$ is a Hopf algebra of Jacobi diagrams on a circular skeleton.

3.1 The Poincaré-Birkhoff-Witt theorem

By forming a Poincaré-Birkhoff-Witt (or PBW) basis for a Lie algebra $\mathfrak{g}$, it is possible to construct an isomorphism to $\mathfrak{U}(\mathfrak{g})$ (taken from [2]).

Theorem 3.1.1. [POINCARÉ, BIRKHOFF, WITT]

Let $\mathfrak{g}$ be a Lie algebra with a totally ordered basis B . Then the universal enveloping algebra $\mathfrak{U}(\mathfrak{g})$ has a basis B' given by

$$B' = \{b_1 b_2 \cdots b_\ell \mid b_i \in B \text{ and } b_i \leq b_{i+1} \forall i\}$$

■

The proof is given in full in [2], and omitted here. This totally ordered basis is termed the PBW basis of $\mathfrak{g}$. It is an immediate consequence that there exists an isomorphism

$$\phi : \mathfrak{g} \rightarrow \mathfrak{U}(\mathfrak{g})$$

The universal enveloping algebra was used to construct $\mathcal{A}^5$, and an isomorphism ψ was described between $\mathfrak{g}$ and $\mathfrak{S}(\mathfrak{g})$ to construct $\mathcal{B}^6$. It follows that the below diagram commutes.

$$\begin{array}{ccc} & \mathfrak{g} & \\ \psi \swarrow & & \searrow \phi \\ \mathfrak{S}(\mathfrak{g}) & \xrightarrow{\chi} & \mathfrak{U}(\mathfrak{g}) \end{array}$$

Equivalently,

Corollary 3.1.2. Let $\mathfrak{g}$ be a Lie algebra, $\mathfrak{U}(\mathfrak{g})$ its universal enveloping algebra, and $\mathfrak{S}(\mathfrak{g})$ its symmetric algebra. Then there exists an isomorphism

$$\chi : \mathfrak{S}(\mathfrak{g}) \rightarrow \mathfrak{U}(\mathfrak{g})$$

where $\chi = \phi \circ \psi^{-1}$.

The final part of this section concerns the diagrammatisation of χ .

3.2 Diagrammatisation

Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be a basis of $\mathfrak{S}(\mathfrak{g})$, constructed by the PBW basis of $\mathfrak{g}$ from (3.1.1). For an element in $\mathfrak{S}(\mathfrak{g})$ generated by $\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_k}$, we would like to diagrammatises the function χ that transforms it through

$$\chi : (\mathbf{e}_{i_1}, \dots, \mathbf{e}_{i_k}) \mapsto \sum_{\sigma \in \mathfrak{S}_k} \mathbf{e}_{\sigma(i_1)} \otimes \cdots \otimes \mathbf{e}_{\sigma(i_k)}$$

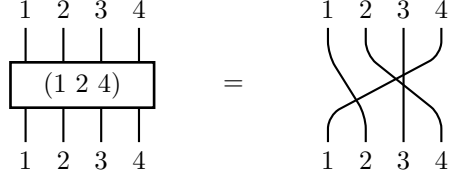
Definition 3.2.1. Given a permutation $\sigma \in \mathfrak{S}_n$, define the permutation σ as a diagram to be

$$\begin{array}{c} 1 \cdots n \\ | \quad | \\ \boxed{\sigma} \\ | \quad | \\ 1 \cdots n \end{array}$$

where the i th strand at the top of the box denoted σ is connected to the $\sigma(i)$ th strand at the bottom of the box. For example,

⁵see [3], Ch. 13, 14

⁶see [4] Sec. 3



Definition 3.2.2. Let $D \in \mathcal{D}_m^o$ be a basis element. Define $\mu(D)$ to be the number of univalent vertices of D . Fix an ordering of these univalent vertices of D , from 1 to $\mu(D)$. Define the map $\chi_m : \mathcal{D}_m^o \rightarrow \mathcal{D}_m$ by

$$\chi_m : D \mapsto \frac{1}{\mu(D)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \underbrace{\left(\begin{array}{c} \text{Diagram of } D \\ \vdots \\ \sigma \\ \vdots \end{array} \right)}_{\mu(D)}$$

where the ellipsis represents all the strands from 1 to m_D , and they are connected accordingly to the univalent vertices in D . So $\chi_m(D)$ represents an average of the sum of all ways of placing the univalent vertices of D on a skeleton, equivalently $\mathbb{S}^1$.

Proposition 3.2.3. There is a linear map

$$\begin{aligned} \bar{\chi}_m : \mathfrak{B}_m &\rightarrow \mathfrak{U}_m \\ [D]_{\mathfrak{B}_m} &\mapsto [\chi(D)]_{\mathfrak{U}_m} \end{aligned}$$

Proof: To map $\bar{\chi}_m$ is well-defined if it is independent of the choice of representative from $[D]_{\mathfrak{B}_m}$. So let $D' \stackrel{\mathfrak{B}}{\sim} D$. Then

$$\chi_m(D) = \frac{1}{\mu(D)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \underbrace{\left(\begin{array}{c} \text{Diagram of } D \\ \vdots \\ \sigma \\ \vdots \end{array} \right)}_{\mu(D)}$$

As $D' \stackrel{\mathfrak{B}}{\sim} D$, there is a finite sequence of AS and IHX moves, which, when applied to D , give D' . As the STU relation implies both the AS and IHX relations⁷, applying the AS and IHX relations to each term of $\chi_m(D)$, it follows that

$$\chi_m(D) \stackrel{\mathfrak{B}}{\sim} \frac{1}{\mu(D)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \underbrace{\left(\begin{array}{c} \text{Diagram of } D' \\ \vdots \\ \sigma \\ \vdots \end{array} \right)}_{\mu(D)} = \frac{1}{\mu(D')!} \sum_{\sigma \in \mathfrak{S}_{\mu(D')}} \underbrace{\left(\begin{array}{c} \text{Diagram of } D' \\ \vdots \\ \sigma \\ \vdots \end{array} \right)}_{\mu(D')} = \chi_m(D')$$

⁷see [3], Ch. 13

Above, $m_D = m_{D'}$ as $D, D' \in \mathfrak{B}_m$. Hence $\chi_m(D) \stackrel{\mathfrak{B}}{\sim} \chi_m(D')$, and $\bar{\chi}_m$ is well-defined.

For linearity, fix $\lambda, \lambda' \in \mathbb{C}$ and $D, D' \in \mathfrak{B}_m$, for which

$$\begin{aligned} \bar{\chi}_m(\lambda[D]_{\mathfrak{B}_m} + \lambda'[D']_{\mathfrak{B}_m}) &= \bar{\chi}_m([\lambda D + \lambda' D']_{\mathfrak{B}_m}) \\ &= [\chi_m(\lambda D + \lambda' D')]_{\mathfrak{A}_m} \\ &= [\lambda \chi_m(D) + \lambda' \chi_m(D')]_{\mathfrak{A}_m} & (*) \\ &= \lambda[\chi_m(D)]_{\mathfrak{A}_m} + \lambda'[\chi_m(D')]_{\mathfrak{A}_m} & (**) \\ &= \lambda \bar{\chi}_m([D]_{\mathfrak{B}_m}) + \lambda' \bar{\chi}_m([D']_{\mathfrak{B}_m}) \end{aligned}$$

Line $(*)$ follows from linearity of χ_m , and $(**)$ follows from the linearity of $[\cdot]_{\mathfrak{A}_m}$, which comes from the definition of a quotient space. Hence $\bar{\chi}_m$ is linear.

This completes the proof. ■

Now we generate the map χ by extending χ_m linearly over $\mathcal{D}^o$, with

$$(\chi : \mathcal{D}^o \rightarrow \mathcal{D}) : \bigoplus_{m \geq 0} (\chi_m : \mathcal{D}_m^o \rightarrow \mathcal{D}_m)$$

Theorem 3.2.4. There is a grade-preserving linear map $\chi : \mathfrak{B} \rightarrow \mathfrak{A}$ given by

$$\chi := \bigoplus_{m \geq 0} \chi_m$$

Proof: Grade-preservation follows as the direct sum does not change the degree of diagrams. ■

Lemma 3.2.5. There is a co-algebra morphism $\bar{\chi} : \mathfrak{B} \rightarrow \mathfrak{A}$ given by

$$\bar{\chi} := \bigoplus_{m \geq 0} \bar{\chi}_m$$

Proof: The result follows from $\chi : \mathcal{D}_m^o \rightarrow \mathcal{D}_m$ being a co-algebra map, or equivalently, that for $D \in \mathcal{D}_m^0$,

$$\Delta(\chi(D)) = (\chi \otimes \chi)(\Delta(D)) \tag{3.1}$$

Without loss of generality, we assume that D is a basis element of $\mathcal{D}_m^o$. Expanding the right-hand side,

$$\begin{aligned} (\chi \otimes \chi)(\Delta(D)) &= (\chi \otimes \chi) \left(\sum_{D_1 \sqcup D_2 = D} D_1 \otimes D_2 \right) \\ &= \sum_{D_1 \sqcup D_2 = D} \chi(D_1) \otimes \chi(D_2) \\ &= \sum_{D_1 \sqcup D_2 = D} \left(\frac{1}{\mu(D_1)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D_1)}} \underbrace{\left(\begin{array}{c} \text{Diagram of } D_1 \text{ with } \sigma \end{array} \right)}_{\mu(D_1)} \otimes \frac{1}{\mu(D_2)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D_2)}} \underbrace{\left(\begin{array}{c} \text{Diagram of } D_2 \text{ with } \sigma \end{array} \right)}_{\mu(D_2)} \right) \end{aligned}$$

Expanding the left-hand side and applying Δ to $\mathcal{A}$, following the definition in (2.3.1), yields

$$\begin{aligned}
\Delta(\chi(D)) &= \Delta \left(\frac{1}{\mu(D)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \underbrace{\left(\begin{array}{c} \leftarrow \\ \boxed{D} \\ \vdots \\ \boxed{\sigma} \\ \vdots \end{array} \right)}_{\mu(D)} \right) \\
&= \frac{1}{\mu(D)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \left(\sum_{X \subset Y} \underbrace{\left(\begin{array}{c} \leftarrow \\ \boxed{D_X} \\ \vdots \\ \boxed{\sigma|_X} \\ \vdots \end{array} \right)}_{\mu(D_X)} \otimes \underbrace{\left(\begin{array}{c} \leftarrow \\ \boxed{D_{Y-X}} \\ \vdots \\ \boxed{\sigma|_{Y-X}} \\ \vdots \end{array} \right)}_{\mu(D_{Y-X})} \right) \\
&= \sum_{X \subset Y} \left(\frac{1}{\mu(D)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \underbrace{\left(\begin{array}{c} \leftarrow \\ \boxed{D_X} \\ \vdots \\ \boxed{\sigma|_X} \\ \vdots \end{array} \right)}_{\mu(D_X)} \otimes \underbrace{\left(\begin{array}{c} \leftarrow \\ \boxed{D_{Y-X}} \\ \vdots \\ \boxed{\sigma|_{Y-X}} \\ \vdots \end{array} \right)}_{\mu(D_{Y-X})} \right) \tag{3.2}
\end{aligned}$$

The switching of sums is allowed, as both are finite. Note the following identity for operators, which is an interchange of sums, for which we need to fix $X \subset Y$. The factor of $\binom{\mu(D_Y)}{\mu(D_X)}$ appears as there are multiple ways to connect $\mu(D_X)$ strands in $\mu(D_Y)$ positions. Here note that $\mu(D) = \mu(D_Y)$ and $\mu(D_Y) - \mu(D_X) = \mu(D_{Y-X})$.

$$\begin{aligned}
\sum_{\sigma \in \mathfrak{S}_{\mu(D)}} \sigma|_X \otimes \sigma|_{Y-X} &= \binom{\mu(D_Y)}{\mu(D_X)} \left(\sum_{\sigma \in \mathfrak{S}_{\mu(D_X)}} \sigma \right) \otimes \left(\sum_{\sigma \in \mathfrak{S}_{\mu(D_{Y-X})}} \sigma \right) \\
&= \mu(D_Y)! \left(\frac{1}{\mu(D_X)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D_X)}} \sigma \right) \otimes \left(\frac{1}{\mu(D_{Y-X})!} \sum_{\sigma \in \mathfrak{S}_{\mu(D_{Y-X})}} \sigma \right)
\end{aligned}$$

Above $\sigma|_X$ represents the map $\sigma \in \mathfrak{S}_{\mu(D)}$ restricted to the strands coming from D_X . Apply this identity

to continue from (3.2) with

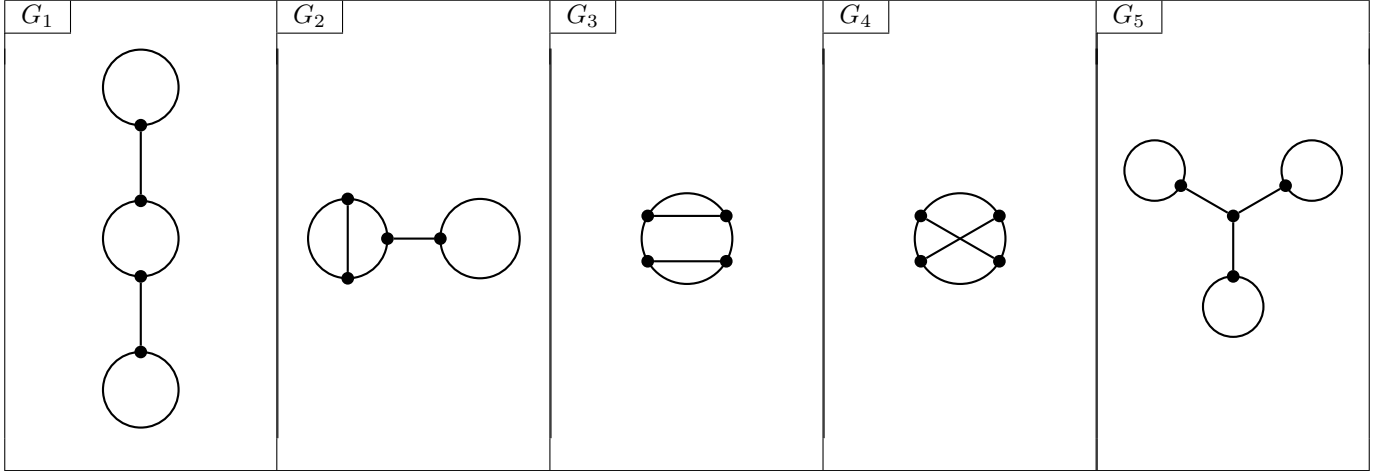
$$\Delta(\chi(D)) = \sum_{X \subset Y} \left(\frac{1}{\mu(D_X)!} \sum_{\sigma \in \mathfrak{S}_{\mu(D_X)}} \underbrace{\left(\begin{array}{c} \boxed{D_X} \\ \vdots \\ \boxed{\sigma} \\ \vdots \end{array} \right)}_{\mu(D_X)} \otimes \frac{1}{\mu(D_{Y-X})!} \sum_{\sigma \in \mathfrak{S}_{\mu(D_{Y-X})}} \underbrace{\left(\begin{array}{c} \boxed{D_{Y-X}} \\ \vdots \\ \boxed{\sigma} \\ \vdots \end{array} \right)}_{\mu(D_{Y-X})} \right)$$

Finally observe that for every pair D_1, D_2 with $D_1 \sqcup D_2 = D$, there is a unique pair $X, Y - X$ for $X \subset Y$ such that $D_X \sqcup D_{Y-X} = D$. Hence both sides of (3.1) are equal. $\blacksquare$

This establishes the desired relationship between $\mathfrak{A}$ and $\mathfrak{B}$.

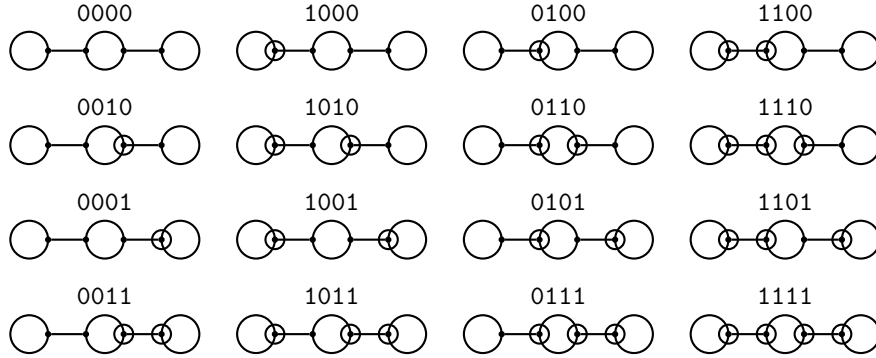
A Construction of $\mathcal{B}_2$

In Section 1.1.1, we found the oonnected diagrams in $\mathcal{D}_2^o$ with 6 edges, or cubic graphs with 4 vertices and 6 edges, to be:



We now examine their symmetry in $\mathcal{B}_2$ to conclude that they reduce to a single instance of G_4 in $\mathcal{B}_2$. When a node is oriented clockwise, a simple circle around it will be the indicator.

G_1 . There are 16 ways that G_1 embeds in D_2^o . Each is assigned a binary string.



Apply rotations and diagram manipulations for identifications among the diagrams.

Rotation of 180° about central vertical axis:

$$\begin{aligned}
 0000 &= 1111 & 1000 &= 1110 \\
 0100 &= 1101 & 0010 &= 1011 \\
 0110 &= 1001 & 0001 &= 0111
 \end{aligned}$$

Rotation of 180° in plane about geometric center:

$$\begin{aligned}
 1000 &= 0001 & 0100 &= 0010 \\
 1100 &= 0011 & 1010 &= 0101 \\
 1110 &= 0111 & 1101 &= 1011
 \end{aligned}$$

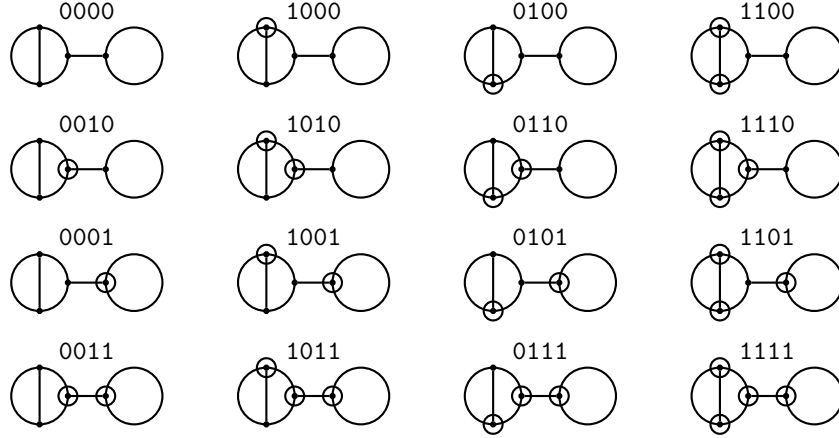
$$\begin{aligned}
 \text{Diagram 0001} &= \text{Diagram 0010} = \text{Diagram 0101} = \text{Diagram 0100} \implies 0001 = 0000 (*) \\
 \text{Diagram 1001} &= \text{Diagram 1010} = \text{Diagram 1011} = \text{Diagram 1101} \implies 1001 = 0000
 \end{aligned}$$

The identity $(*)$ implies $0011 = 0010$ and $0101 = 0100$. Therefore:

$$\begin{aligned}
0000 &= 1111 = 0001 = 0111 = 1000 = 1110 = 1001 = 0110 \\
0010 &= 1011 = 0100 = 1101 = 0011 = 1100 = 0101 = 1010
\end{aligned}$$

Hence G_1 generates two distinguishable diagrams in $\mathcal{D}_2^0$, 0000 and 0010.

G_2 . There are 16 ways that G_2 embeds in D_2^0 . Each is assigned a binary string.



Apply rotations and diagram manipulations for identifications among the diagrams.

Rotation of 180° about central horizontal axis:

$$\begin{aligned}
0000 &= 1111 & 1000 &= 1011 & 0100 &= 0111 & 0010 &= 1101 \\
0001 &= 1110 & 1100 &= 0011 & 1010 &= 1001 & 0101 &= 0110
\end{aligned}$$

$$\begin{array}{c} \text{Diagram with node on left vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on right vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on horizontal line} \end{array} \Rightarrow 1100 = 0000 (*)$$

$$\begin{array}{c} \text{Diagram with node on left vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on right vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on horizontal line} \end{array} = \begin{array}{c} \text{Diagram with node on left vertical line} \end{array} \Rightarrow 0001 = 0000 (**)$$

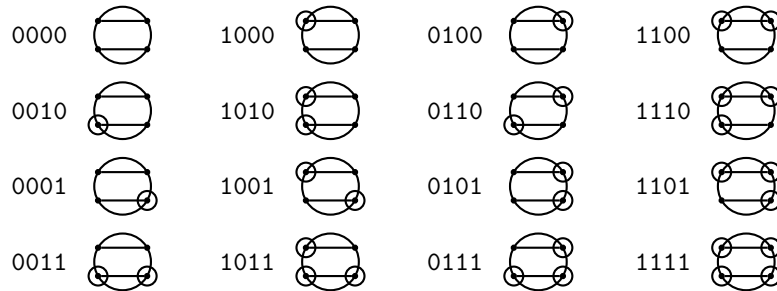
$$\begin{array}{c} \text{Diagram with node on left vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on right vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on horizontal line} \end{array} = \begin{array}{c} \text{Diagram with node on left vertical line} \end{array} = \begin{array}{c} \text{Diagram with node on left vertical line} \end{array} \Rightarrow 0010 = 0000 (***)$$

Identity (*) was applied to calculate identity (***) in the last equality. The identity (*) implies $0100 = 1000$. The identity (**) implies $0111 = 0110$. The identity (***) implies $1011 = 1001$. Therefore:

$$\begin{aligned}
0000 &= 1111 = 1100 = 0011 = 0001 = 1110 = 0010 = 1101 \\
0100 &= 0111 = 0110 = 0101 = 1000 = 1011 = 1001 = 1010
\end{aligned}$$

Hence G_2 generates two distinguishable diagrams in $\mathcal{D}_2^0$, 0000 and 0100.

G_3 . There are 16 ways that G_3 embeds in D_2^0 . Each is assigned a binary string.



Apply rotations and diagram manipulations for identifications among the diagrams.

Rotation of 180° about central horizontal axis:

$$\begin{aligned} 0000 &= 1111 & 1000 &= 1101 \\ 0100 &= 1110 & 0010 &= 0111 \\ 0001 &= 1011 & 1010 &= 0101 \end{aligned}$$

Rotation of 180° about central vertical axis:

$$\begin{aligned} 1000 &= 1011 & 0100 &= 0111 \\ 0010 &= 1110 & 0001 &= 1101 \\ 1100 &= 0011 \end{aligned}$$

$$\begin{aligned} \text{Diagram 1} &= \text{Diagram 2} = \text{Diagram 3} = \text{Diagram 4} = \text{Diagram 5} \Rightarrow 0001 = 0010 (*) \\ \text{Diagram 6} &= \text{Diagram 7} = \text{Diagram 8} \Rightarrow 0011 = 0000 (**) \end{aligned}$$

The identity (*) implies $1111 = 1100$ and $1101 = 1110$ and $1010 = 1001$ and $1010 = 0110$. The identity (**) implies $1100 = 0000$. Therefore:

$$\begin{aligned} 0000 &= 1111 = 0011 = 1100 \\ 0001 &= 1011 = 1101 = 1110 = 0100 = 0111 = 1000 = 0010 \\ 0101 &= 1010 = 1001 = 0110 \end{aligned}$$

Hence G_3 generates three distinguishable diagrams in $\mathcal{D}_2^0$, 0000, 0001 and 0101.

G_4 . There are 16 ways that G_4 embeds in D_2^0 . Each is assigned a binary string.

0000		1000		0100		1100	
0010		1010		0110		1110	
0001		1001		0101		1101	
0011		1011		0111		1111	

Apply rotations and diagram manipulations for identifications among the diagrams.

Rotation of 90° in plane about geometric center

$$\begin{aligned} 1000 &= 0100 & 0100 &= 0001 \\ 0001 &= 0010 & 1100 &= 0101 \\ 0101 &= 0011 & 0011 &= 1010 \\ 0110 &= 1001 & 1110 &= 1101 \\ 1101 &= 0111 & 0111 &= 1011 \end{aligned}$$

Rotation of 180° about central vertical axis:

$$\begin{aligned} 0000 &= 1111 & 1000 &= 1011 \\ 0100 &= 0111 & 0010 &= 1110 \\ 0001 &= 1101 & 1100 &= 0011 \end{aligned}$$

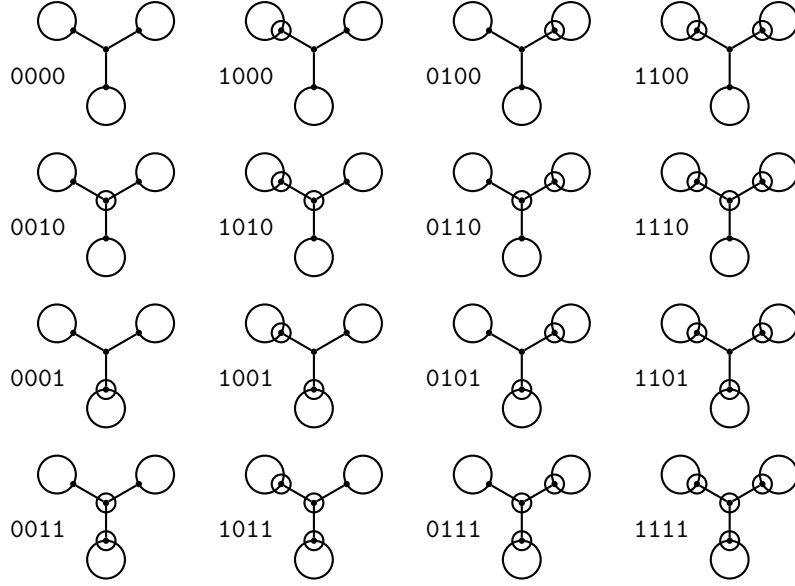
$$\text{Diagram 1} = \text{Diagram 2} = \text{Diagram 3} = \text{Diagram 4} = \text{Diagram 5} \Rightarrow 0000 = 1100 (*)$$

The identity (*) implies $0000 = 0011$ and $0110 = 1001$. Therefore:

$$\begin{aligned} 0000 &= 1111 = 0011 = 1100 = 1010 = 0101 \\ 0001 &= 0010 = 0100 = 1000 = 1011 = 0111 = 1110 = 1101 \\ 0110 &= 1001 \end{aligned}$$

Hence G_4 generates three distinguishable diagrams in $\mathcal{D}_2^0$, 0000, 0001 and 0110.

G_5 . There are 16 ways that G_5 embeds in D_2^0 . Each is assigned a binary string.



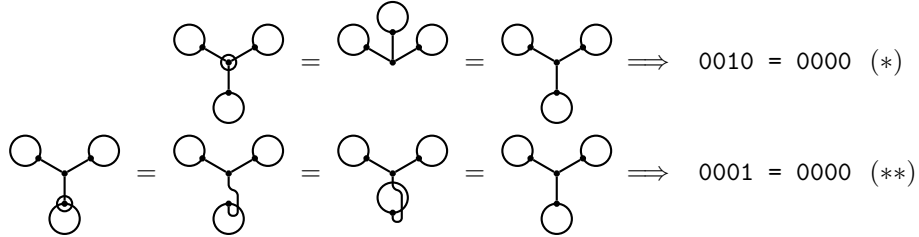
Apply rotations and diagram manipulations for identifications among the diagrams.

Rotation of 120° in plane about geometric center

$$\begin{aligned} 1000 &= 0100 & 0100 &= 0001 \\ 1010 &= 0110 & 0110 &= 0011 \\ 1100 &= 0101 & 0101 &= 1001 \\ 1110 &= 0111 & 0111 &= 1110 \end{aligned}$$

Rotation of 180° about central vertical axis:

$$\begin{aligned} 0000 &= 1111 & 1000 &= 1011 \\ 0100 &= 0111 & 0010 &= 1101 \\ 0001 &= 1110 & 1100 &= 0011 \\ 1001 &= 1010 & 0101 &= 0110 \end{aligned}$$

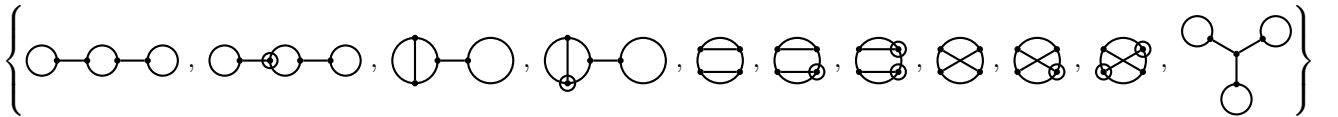


Identity (*) implies $1000 = 1010$. Identity (**) implies $1101 = 0000$. Therefore:

$$\begin{aligned} 0000 &= 1111 = 0001 = 0100 = 1000 = 1110 = 0111 = 1011 \\ &= 0010 = 1010 = 0110 = 0011 = 1100 = 0101 = 1001 = 1101 \end{aligned}$$

Hence G_5 generates one distinguishable diagram in $\mathcal{D}_2^0$, 0000.

Now we apply the IHX relation to determine dependence in $\mathcal{B}_2$ among the following diagrams:



Remark A.1. By a straightforward application of the AS relation, a diagram with n clockwise-oriented vertices is equal to that diagram with any other m clockwise oriented vertices, for $n \equiv m \pmod{2}$. Hence

$$\begin{array}{c} \text{Diagram 1} \end{array} \stackrel{\mathcal{B}}{\sim} \begin{array}{c} \text{Diagram 2} \end{array} \quad \begin{array}{c} \text{Diagram 3} \end{array} \stackrel{\mathcal{B}}{\sim} \begin{array}{c} \text{Diagram 4} \end{array}$$

Since $-1 \in \mathbb{C}$, diagrams in $\mathcal{B}_2$ from the same Jacobi diagram are equivalent, so

$$\begin{array}{c} \text{Diagram 5} \end{array} \stackrel{\mathcal{B}}{\sim} \begin{array}{c} \text{Diagram 6} \end{array} \quad \begin{array}{c} \text{Diagram 7} \end{array} \stackrel{\mathcal{B}}{\sim} \begin{array}{c} \text{Diagram 8} \end{array}$$

By the observation in (1.1.3), also from the AS relation,

$$\begin{array}{c} \text{Diagram 9} \end{array} \stackrel{\mathcal{B}}{\sim} 0 \quad \begin{array}{c} \text{Diagram 10} \end{array} \stackrel{\mathcal{B}}{\sim} 0 \quad \begin{array}{c} \text{Diagram 11} \end{array} \stackrel{\mathcal{B}}{\sim} 0 \quad \begin{array}{c} \text{Diagram 12} \end{array} \stackrel{\mathcal{B}}{\sim} 0 \quad \begin{array}{c} \text{Diagram 13} \end{array} \stackrel{\mathcal{B}}{\sim} 0$$

Now we selectively apply the IHX relation.

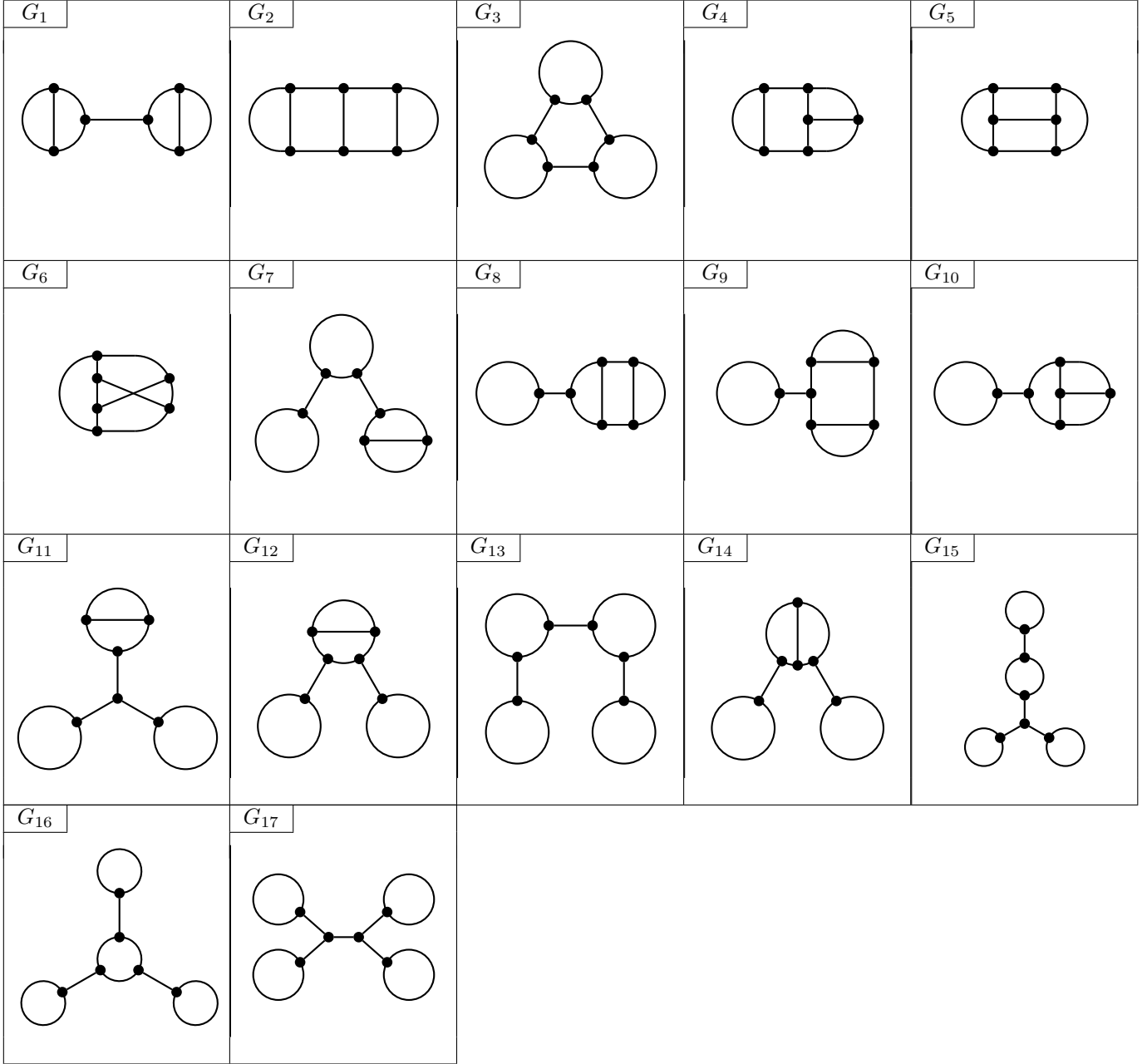
$$\begin{aligned} \begin{array}{c} \text{Diagram 14} \end{array} &= \begin{array}{c} \text{Diagram 15} \end{array} - \begin{array}{c} \text{Diagram 16} \end{array} \\ &= \begin{array}{c} \text{Diagram 17} \end{array} - \begin{array}{c} \text{Diagram 18} \end{array} \\ \Rightarrow 2 \begin{array}{c} \text{Diagram 19} \end{array} &= \begin{array}{c} \text{Diagram 20} \end{array} \end{aligned}$$

Therefore the 80 unique vertex-oriented cubic graphs embed as 11 unique open Jacobi diagrams, and as a single Jacobi diagram in $\mathcal{B}_2$,

$$\begin{array}{c} \text{Diagram 21} \end{array} = \begin{array}{c} \text{Diagram 22} \end{array}$$

B Construction of $\mathcal{B}_3$

Using the programs described in Appendix C, we find the set of cubic graphs with 6 vertices and 9 edges to be:



Applying (A.1) implies that at most 6 distinct diagrams among the ones above with oriented vertices, embed into $\mathcal{B}_3$. This follows as $G_7, \dots, G_{17}$ are all equivalent to zero, and only one diagram among the vertex-oriented diagrams of each of $G_1, \dots, G_6$ is distinct in $\mathcal{B}_3$.

Apply the IHX relation, to find more identifications.

$$G_1 = \text{Diagram 1} = \text{Diagram 2} - \text{Diagram 3} = G_2 - (-1)^2 G_2 = 0 \quad (\text{B.1})$$

$$G_3 = \text{Diagram 4} = \text{Diagram 5} - \text{Diagram 6} = G_4 - (-1)G_4 = 2G_4 \quad (\text{B.2})$$

$$G_5 = \text{Diagram 7} = \text{Diagram 8} - \text{Diagram 9} = G_5 - (-1)^2 G_6 \implies G_6 = 0 \quad (\text{B.3})$$

$$G_2 = \text{Diagram 10} = \text{Diagram 11} - \text{Diagram 12} = G_4 - (-1)G_4 = 2G_4 \quad (\text{B.4})$$

$$G_4 = \text{Diagram 13} = \text{Diagram 14} - \text{Diagram 15} = G_5 - (-1)G_5 = 2G_5 \quad (\text{B.5})$$

Hence $G_1 = G_6 = 0$, and G_2, G_3, G_4 may all be expressed in terms of G_5 . Therefore the 1088 unique vertex-oriented cubic graphs embed as a single Jacobi diagram in $\mathcal{B}_3$,

$$\text{Diagram 16} = \text{Diagram 17}$$

Remark B.1. The first equation (B.1) above demonstrates a general rule, that if a diagram D has a bridge, then $D \stackrel{\mathcal{B}}{\sim} 0$, by letting the bridge be the ‘I’ in the IHX relation. In fact, the observation (1.1.3), that a

diagram with a loop evaluates to zero by the AS relation, is a special case of the described property, as a diagram with a loop necessarily has a bridge connecting the loop to the rest of the diagram.

The second, fourth and last equations (B.2), (B.4), and (B.5), demonstrate another general rule, that if a diagram has a cut of size two, then it may be expressed in terms of another diagram D' , with $D \stackrel{\mathcal{B}}{\sim} 2D'$, by letting one of the edges in the cut of size two be the ‘I’ in the IHX relation. Note this only holds if the edges in the cut are not parallel.

From these two observations it follows that, as G_1 has a bridge, and G_2 , G_3 , and G_4 all have cuts of size 2, that $\mathcal{B}_3$ has at most 3 distinct elements.

C Graph generation

Two approaches are used, one to determine the actual diagrams in $\mathcal{B}_m$, and the second to find an upper bound on the number of distinct diagrams in $\mathcal{B}_m$.

C.1 Method

The first approach used is naive, but fast enough for small degrees (1,2,3). All possible incidence matrices of trivalent graphs, with parallel edges and loops allowed, are generated with Sage⁸ software (in the Python language), then the list is checked for isomorphisms with Mathematica software. In essence:

- Compile the Sage and Mathematica helper functions
- In the main Sage and Mathematica sequences, set the variable `targetdegree` to be the desired degree
- Run the main Sage sequence, then the main Mathematica sequence

This will display, in Mathematica, all non-isomorphic graphs with parallel edges and loops of degree `targetdegree`.

The files used in this approach are available online, at www.jlazovskis.com/scripts/HA2012/

helper functions (Sage):	<code>s1-hfun.py</code>
main sequence (Sage):	<code>s1-mseq.py</code>
helper functions (Mathematica):	<code>m1-hfun.nb</code>
main sequence (Mathematica):	<code>m1-mseq.nb</code>

The dimensions of the space of diagrams produced for $\mathcal{B}_1$, $\mathcal{B}_2$, $\mathcal{B}_3$ are 2, 5, 17. This does not give a good indication of the speed at which the dimension increases, as the next few terms are 71, 388, 2592, 21096. The dimension up to 16 is given by sequence A005967 at the OEIS⁹.

As described previously, many of the diagrams calculated in the above approach either reduce to zero or are redundant. By computing only loopless graphs in a second approach, we find an upper bound on the number of distinct diagrams in $\mathcal{B}_m$. The program for generating these graphs uses Brandon McKay's graph generating program `nauty`¹⁰. In essence:

- Compile the Sage and Mathematica helper functions
- In the main Shell script, Sage, and Mathematica sequences, set the variable `targetdegree`
- Run the main Shell script sequence, then the Sage sequence, and finally the Mathematica sequence

This will display, in Mathematica, all non-isomorphic loopless graphs with parallel edges of degree `targetdegree`.

The files used in this approach are available online as above.

main sequence (Shell script):	<code>sh2-mseq.sh</code>
helper functions (Sage):	<code>s2-hfun.py</code>
main sequence (Sage):	<code>s2-mseq.py</code>
helper functions (Mathematica):	<code>m2-hfun.nb</code>
main sequence (Mathematica):	<code>m2-mseq.nb</code>

This approach runs reasonably fast for degrees 1 to 5, but at degree = 6, as `nauty` generates more than 7 million graphs, classification in Sage requires a prohibitively large amount of memory. The sequence of degrees for these loopless graphs, up to dimension 12, is given by the OEIS sequence A000421, the first seven terms being 1, 2, 6, 20, 91, 509, 3608, 31856.

⁸available online at www.sagemath.org

⁹the Online Encyclopedia of Integer Sequences, at www.oeis.org, maintained by Neil Sloane

¹⁰available online at cs.anu.edu.au/~bdm/nauty

C.2 Complexity analysis

Here we will only analyze the file at `s1-mseq.py`. For `deg = 1` and `deg = 2`, the program runs in less than a second. For `deg = 3`, it takes about 6.5 minutes for the program to terminate on a personal computer rated at 2.4 GHz. For `deg = 4`, the program did not terminate in 10 hours.

Let k be the size of the primary input for each function.

- Line 9: The function `twos` has running time k
- Line 10: The function `ones` has running time k^2
- Line 11: The sorting function has running time $k \log(k)$
- Line 12: The reverse function has constant running time k
- Line 14: The function `copy.copy` has running time k
- Line 16: The function `Combinations` has running time 2^k
- Line 21: The function `chg` has running time k

The rest of the unmentioned functions have constant running time. In every iteration of the main loop, the list of usable vectors `usel` excludes extraneous vectors that would make $\Delta(G) > 3$ for graphs G generated in the current iteration. However, this only reduces time complexity by a polynomial factor, overshadowed by the exponential time complexity of `Combinations`.

Let n be the value to which `targetdegree` is set to in the main sequence. Then the computational complexity of the main sequence is exponential:

$$\mathcal{O}(n) + \mathcal{O}(n^2) + \mathcal{O}(n^2 \log(n^2)) + \mathcal{O}(n^2) + \mathcal{O}(n) \left(\mathcal{O}(n) + \mathcal{O}(2^{n^2}) + \mathcal{O}(n) (\mathcal{O}(n) + \mathcal{O}(n) + \mathcal{O}(n)) \right) = \mathcal{O} \left(2^{n^2} \right)$$

Hence this approach is fairly slow in terms of graph generation, but enough for our purposes.

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